DATA MINING

PROJECT REPORT

By

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Problem 2 PCA: PCA FH (FT): Primary census abstract for female headed households excluding institutional households (India & States/UTs - District Level), Scheduled tribes - 2011 PCA for Female Headed Household Excluding Institutional Household. The Indian Census has the reputation of being one of the best in the world. The first Census in India was conducted in the year 1872. This was conducted at different points of time in different parts of the country. In 1881 a Census was taken for the entire country simultaneously. Since then, Census has been conducted every ten years, without a break. Thus, the Census of India 2011 was the fifteenth		22

in this unbroken series since 1872, the seventh after independence and the second census of the third millennium and twenty first century. The census has been uninterruptedly continued despite of several adversities like wars, epidemics, natural calamities, political unrest, etc. The Census of India is conducted under the provisions of the Census Act 1948 and the Census Rules, 1990. The Primary Census Abstract which is important publication of 2011 Census gives basic information on Area, Total Number of Households, Total Population, Scheduled Castes, Scheduled Tribes Population, Population in the age group 0-6, Literates, Main Workers and Marginal Workers classified by the four broad industrial categories, namely, (i) Cultivators, (ii) Agricultural Laborers, (iii) Household Industry Workers, and (iv) Other Workers and also non-Workers. The characteristics of the Total Population include Scheduled Castes, Scheduled Tribes, Institutional and Houseless Population and are presented by sex and rural-urban residence. Census 2011 covered 35 States/Union Territories, 640 districts, 5,924 sub-districts, 7,935 Towns and 6,40,867 Villages. The data collected has so many variables thus making it difficult to find useful details without using Data Science Techniques. You are tasked to perform detailed EDA and identify Optimum Principal Components that explains the most variance in data. Use Sklearn only.		
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Problem 1

Clustering
Digital Ads Data

The ads24x7 is a Digital Marketing company which has now got seed funding of $10 Million. They are expanding their wings in Marketing Analytics. They collected data from their Marketing Intelligence team and now wants you (their newly appointed data analyst) to segment type of ads based on the features provided. Use Clustering procedure to segment ads into homogeneous groups.

The following three features are commonly used in digital marketing:

CPM = (Total Campaign Spend / Number of Impressions) * 1,000. Note that the Total Campaign Spend refers to the 'Spend' Column in the dataset and the Number of Impressions refers to the 'Impressions' Column in the dataset.

CPC = Total Cost (spend) / Number of Clicks. Note that the Total Cost (spend) refers to the 'Spend' Column in the dataset and the Number of Clicks refers to the 'Clicks' Column in the dataset.

CTR = Total Measured Clicks / Total Measured Ad Impressions x 100. Note that the Total Measured Clicks refers to the 'Clicks' Column in the dataset and the Total Measured Ad Impressions refers to the 'Impressions' Column in the dataset.

The Data Dictionary and the detailed description of the formulas for CPM, CPC and CTR are given in the sheet 2 of the Clustering Clean ads_data Excel File.

Perform the following in given order:

1.1 Read the data and perform basic analysis such as printing a few rows (head and tail), info, data summary, null values duplicate values, etc.

In this case study, we will examine how we can approach decisions related to ads24x7. After importing the dataset, we check the shape of the data, the types of variables and missing values in the dataset.

First five rows of the dataset are shown below.

	Timestamp	Inventory Type	Ad - Length	Ad-Width	Ad Size	Ad Type	Platform	Device Type	Format	Available_Impressions	Matched_Queries	Impressions	Clicks	Spend
0	2020-9-2-17	Format1	300	250	75000	Inter222	Video	Desktop	Display	1806	325	323	1	0.0
1	2020-9-2-10	Format1	300	250	75000	Inter227	App	Mobile	Video	1780	285	285	1	0.0
2	2020-9-1-22	Format1	300	250	75000	Inter222	Video	Desktop	Display	2727	356	355	1	0.0
3	2020-9-3-20	Format1	300	250	75000	Inter228	Video	Mobile	Video	2430	497	495	1	0.0
4	2020-9-4-15	Format1	300	250	75000	Inter217	Web	Desktop	Video	1218	242	242	1	0.0

Fig.1 Head of the dataset

Last 5 rows of the dataset are shown below.

	Timestamp	InventoryType	Ad - Length	Ad- Width	Ad Size	Ad Type	Platform	Device Type	Format	Available_Impressions	Matched_Queries	Impressions	Clicks
23061	2020-9-13-7	Format5	720	300	216000	Inter220	Web	Mobile	Video	1	1	1	1
23062	2020-11-2-7	Format5	720	300	216000	Inter224	Web	Desktop	Video	3	2	2	1
23063	2020-9-14-22	Format5	720	300	216000	Inter218	App	Mobile	Video	2	1	1	1
23064	2020-11-18-2	Format4	120	600	72000	inter230	Video	Mobile	Video	7	1	1	1
23065	2020-9-14-0	Format5	720	300	216000	Inter221	App	Mobile	Video	2	2	2	1

Fig.2 Tail of the dataset

The dataset comprises of **23066 rows** and **19 columns**. There are some null values in the dataset.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23066 entries, 0 to 23065
Data columns (total 19 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Timestamp              23066 non-null  object
 1   InventoryType          23066 non-null  object
 2   Ad - Length            23066 non-null  int64
 3   Ad- Width              23066 non-null  int64
 4   Ad Size                23066 non-null  int64
 5   Ad Type                23066 non-null  object
 6   Platform               23066 non-null  object
 7   Device Type            23066 non-null  object
 8   Format                 23066 non-null  object
 9   Available_Impressions  23066 non-null  int64
 10  Matched_Queries        23066 non-null  int64
 11  Impressions            23066 non-null  int64
 12  Clicks                 23066 non-null  int64
 13  Spend                  23066 non-null  float64
 14  Fee                    23066 non-null  float64
 15  Revenue                23066 non-null  float64
 16  CTR                    18330 non-null  float64
 17  CPM                    18330 non-null  float64
 18  CPC                    18330 non-null  float64
dtypes: float64(6), int64(7), object(6)
memory usage: 3.3+ MB
```

Fig.3 Information of the dataset

	count	mean	std	min	25%	50%	75%	max
Ad - Length	23066.0	385.16	233.65	120.00	120.00	300.00	720.00	728.00
Ad- Width	23066.0	337.90	203.09	70.00	250.00	300.00	600.00	600.00
Ad Size	23066.0	96674.47	61538.33	33600.00	72000.00	72000.00	84000.00	216000.00
Available_Impressions	23066.0	2432043.67	4742887.76	1.00	33672.25	483771.00	2527711.75	27592861.00
Matched_Queries	23066.0	1295099.14	2512969.86	1.00	18282.50	258087.50	1180700.00	14702025.00
Impressions	23066.0	1241519.52	2429399.96	1.00	7990.50	225290.00	1112428.50	14194774.00
Clicks	23066.0	10678.52	17353.41	1.00	710.00	4425.00	12793.75	143049.00
Spend	23066.0	2706.63	4067.93	0.00	85.18	1425.12	3121.40	26931.87
Fee	23066.0	0.34	0.03	0.21	0.33	0.35	0.35	0.35
Revenue	23066.0	1924.25	3105.24	0.00	55.37	926.34	2091.34	21276.18
CTR	18330.0	0.07	0.08	0.00	0.00	0.08	0.13	1.00
CPM	18330.0	7.67	6.48	0.00	1.71	7.66	12.51	81.56
CPC	18330.0	0.35	0.34	0.00	0.09	0.16	0.57	7.26

Fig.4 Description of the dataset

The **describe method** will provide statistical information of the numerical values in the dataset like the mean, median, max. and min. values etc.

There are **no duplicate values** in the dataset.

1.2 Treat missing values in CPC, CTR and CPM using the formula given.

There are 4736 missing values each in CTR, CPM, CPC columns in the dataset. These missing values are to be imputed by creating a user defined function.

The below mentioned formulas are used for imputation of missing values.

CPM = (Total Campaign Spend / Number of Impressions) * 1,000
CPC = Total Cost (spend) / Number of Clicks
CTR = Total Measured Clicks / Total Measured Ad Impressions x 100

Timestamp 0 InventoryType 0 Ad - Length 0 Ad- Width 0 Ad Size 0 Ad Type 0 Platform 0 Device Type 0 Format 0 Available_Impressions 0 Matched_Queries 0 Impressions 0 Clicks 0 Spend 0 Fee 0 Revenue 0 CTR 4736 CPM 4736 CPC 4736 dtype: int64	Timestamp 0 InventoryType 0 Ad - Length 0 Ad- Width 0 Ad Size 0 Ad Type 0 Platform 0 Device Type 0 Format 0 Available_Impressions 0 Matched_Queries 0 Impressions 0 Clicks 0 Spend 0 Fee 0 Revenue 0 CTR 0 CPM 0 CPC 0 dtype: int64
Before imputing missing values	**After imputing missing values**

Fig. 5 Treating missing values

1.3 Check if there are any outliers. Do you think treating outliers is necessary for K Means clustering? Based on your judgement decide whether to treat outliers and if yes, which method to employ. (As an analyst your judgement may be different from another analyst)

While checking Boxplot for outliers it can be seen that there are significant number of outliers. Now, if the data set contains outliers, there may be data preprocessing or statistical analysis errors. Any outlier in the data set can skew the test results and lead to an erroneous interpretation of the data. The K-means clustering algorithm is sensitive to outliers, because a mean is easily influenced by extreme values. Thus, the removal of outliers is an essential task in the analysis and processing of data.

In this case, we will treat outliers using IQR method.

We have created a separate dataset df_num consisting of numerical variables from the original dataset for further computational analysis.

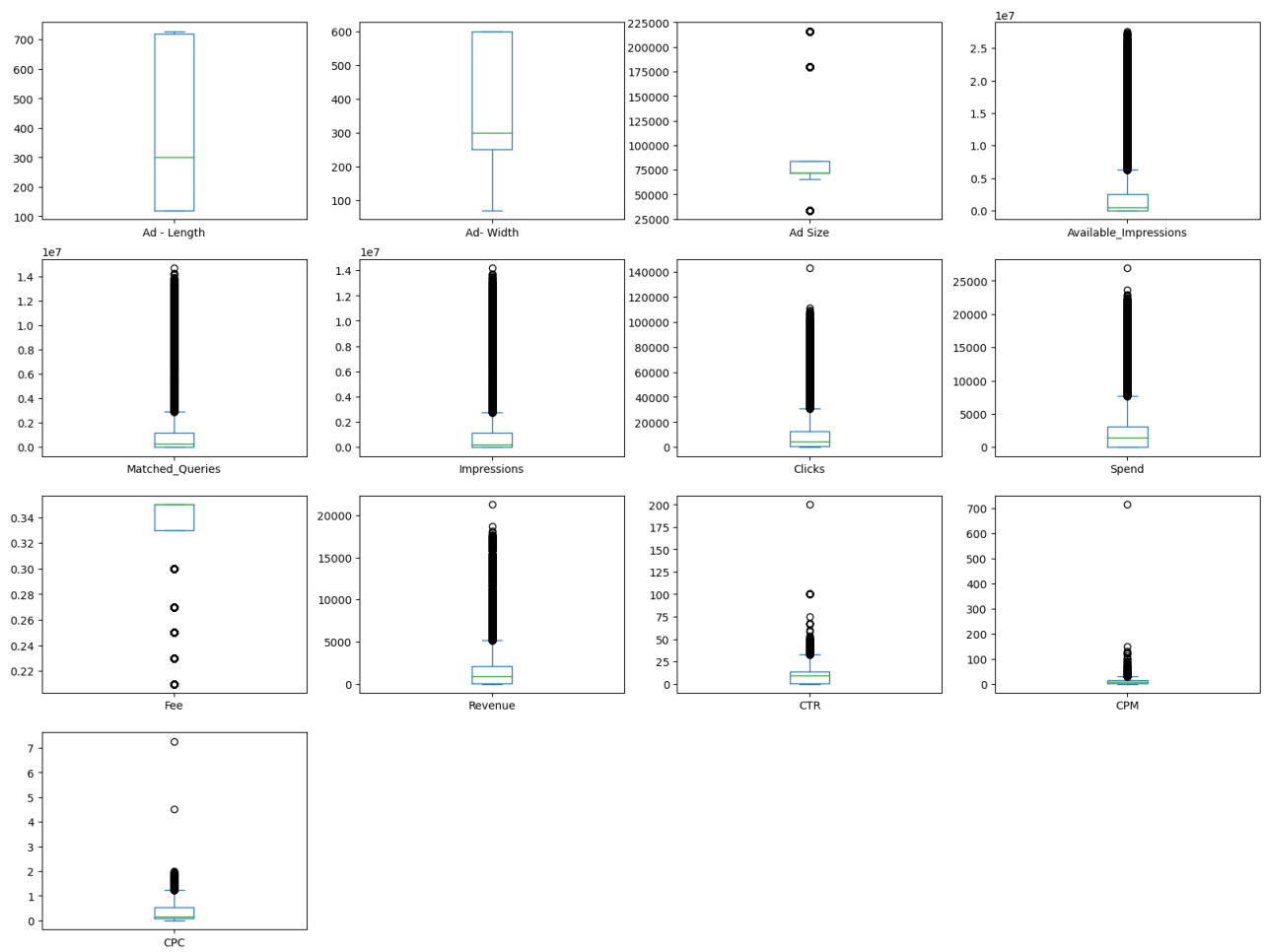


Fig. 6 Box plot before outlier treatment

After outlier treatment, the data points are capped between 5th and 95th percentile value of the distribution.

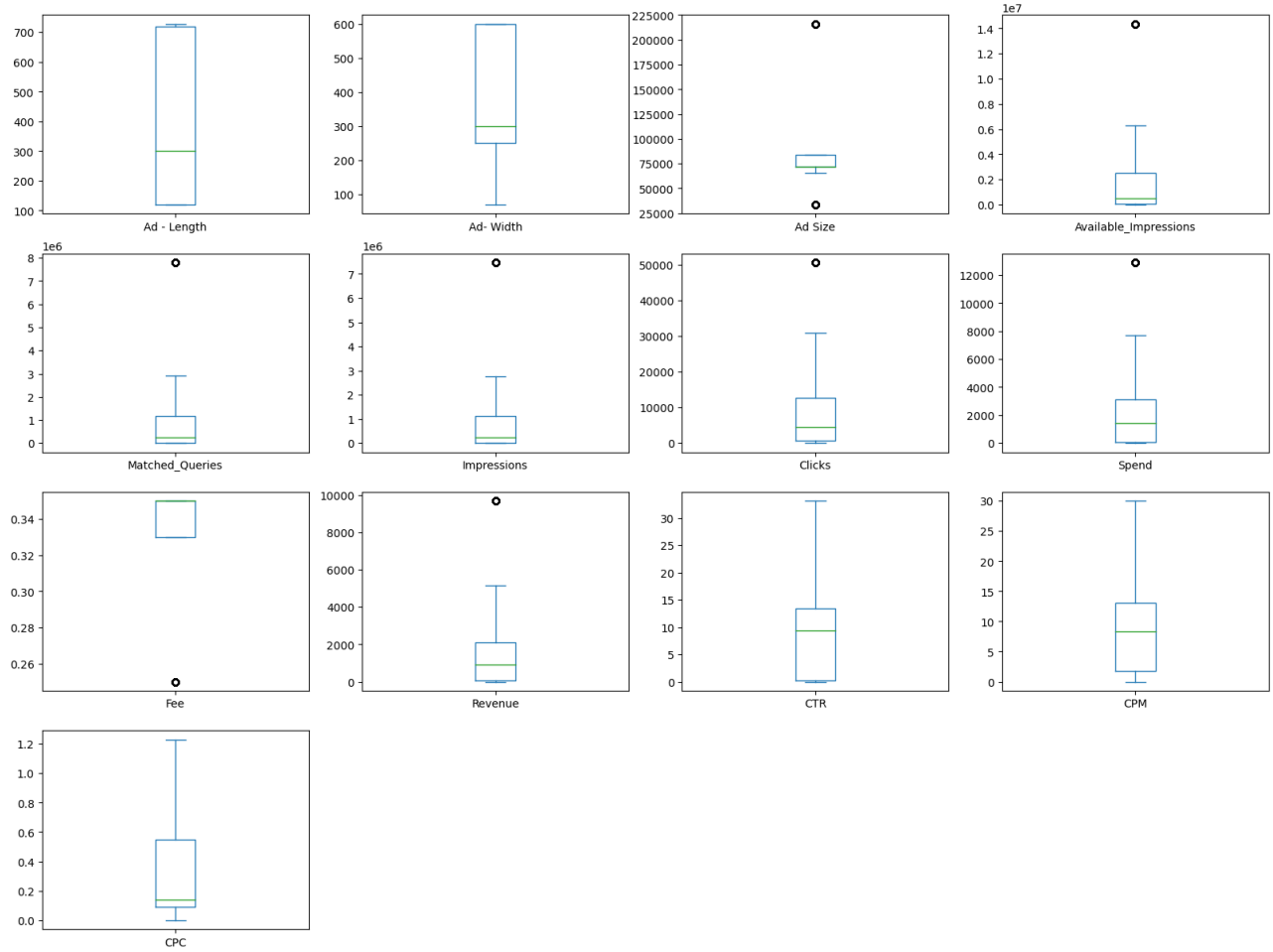


Fig. 7 Box plot after outlier treatment

1.4 Perform z-score scaling and discuss how it affects the speed of the algorithm.

We used scikit-learn's StandardScaler to perform z-score scaling. First 5 rows of scaled data are shown below.

	0	1	2	3	4
Ad - Length	-0.364496	-0.364496	-0.364496	-0.364496	-0.364496
Ad- Width	-0.432797	-0.432797	-0.432797	-0.432797	-0.432797
Ad Size	-0.359227	-0.359227	-0.359227	-0.359227	-0.359227
Available_Impressions	-0.569484	-0.569490	-0.569269	-0.569339	-0.569622
Matched_Queries	-0.567061	-0.567076	-0.567049	-0.566994	-0.567093
Impressions	-0.563943	-0.563958	-0.563931	-0.563875	-0.563975
Clicks	-0.719779	-0.719779	-0.719779	-0.719779	-0.719779
Spend	-0.722776	-0.722776	-0.722776	-0.722776	-0.722776
Fee	0.487214	0.487214	0.487214	0.487214	0.487214
Revenue	-0.676118	-0.676118	-0.676118	-0.676118	-0.676118
CTR	-0.978830	-0.973650	-0.982332	-0.992329	-0.965826
CPM	-1.220346	-1.220346	-1.220346	-1.220346	-1.220346
CPC	-1.083011	-1.083011	-1.083011	-1.083011	-1.083011

Fig. 8 Scaled dataset

Scaling is a fundamental preprocessing step in machine learning that involves adjusting the magnitude of input features. Its importance lies in ensuring that all features contribute equally to the learning process, preventing dominance by features with larger scales. Scaling promotes faster convergence of optimization algorithms, enhances model performance, and facilitates the correct functioning of distance-based algorithms like K-Means. It also aids in feature interpretability, prevents numerical instability issues, and promotes consistent model performance. Overall, scaling is a crucial practice to optimize machine learning models for improved accuracy and stability**.**

1.5 Perform Hierarchical by constructing a Dendrogram using WARD and Euclidean distance and identify the optimum number of clusters.

Using SciPy's cluster hierarchy function, we created the below dendrogram

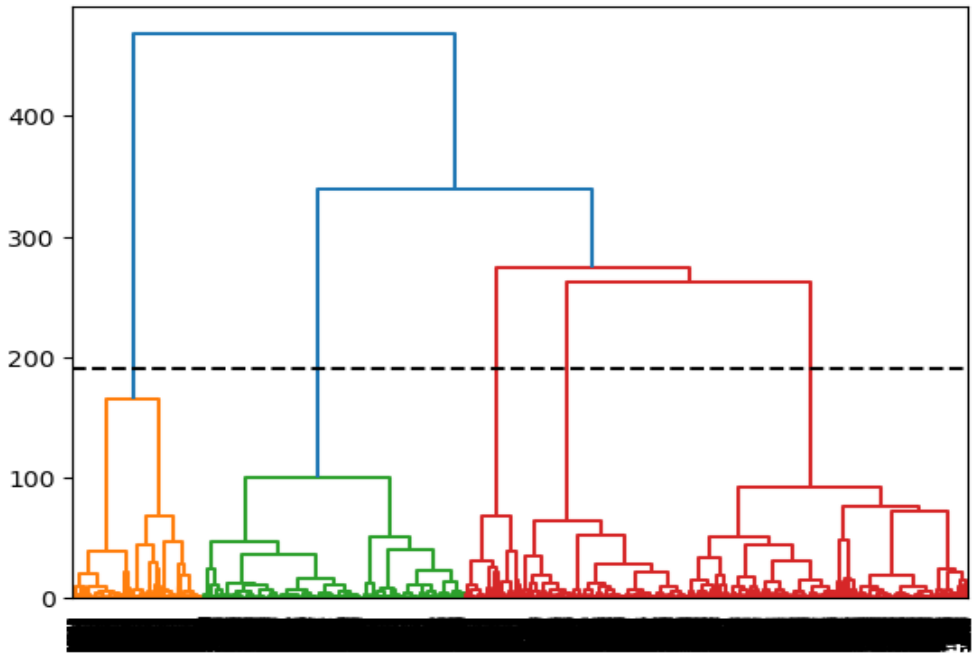


Fig. 9 Dendrogram using WARD and Euclidean distance

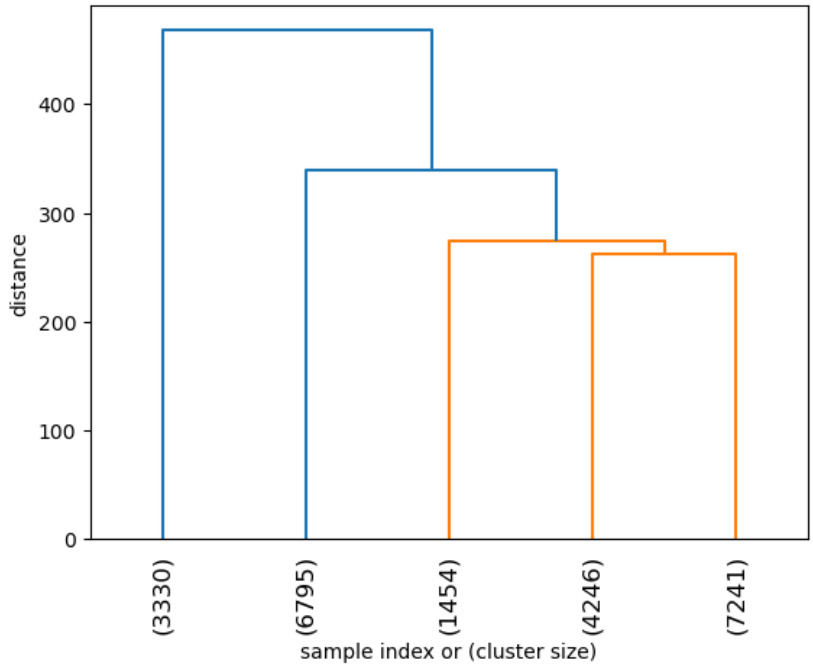


Fig. 10 Hierarchical clustering Dendrogram (truncated)

Building upon the aforementioned reference, we observe that the most extended branch, characterized by its height, corresponds to the blue segment. However, focusing solely on the blue segment would yield only two clusters, which would be unsuitable for business purposes. On the contrary, if we employ segmentation criteria based on the tallest red branches, demarcated by the horizontal black line, we can identify five distinct clusters. In the present project, we opt to utilize the Dendrogram method to identify and work with a total of **five clusters**.

After considering the dendrogram, one may decide the level where the resultant tree needs to be cut. This is another way to say that how close-knit or dispersed should the clusters be. Practical consideration or domain knowledge plays a part in determining the number of clusters.

1.6 Make Elbow plot (up to n=10) and identify optimum number of clusters for k-means algorithm.

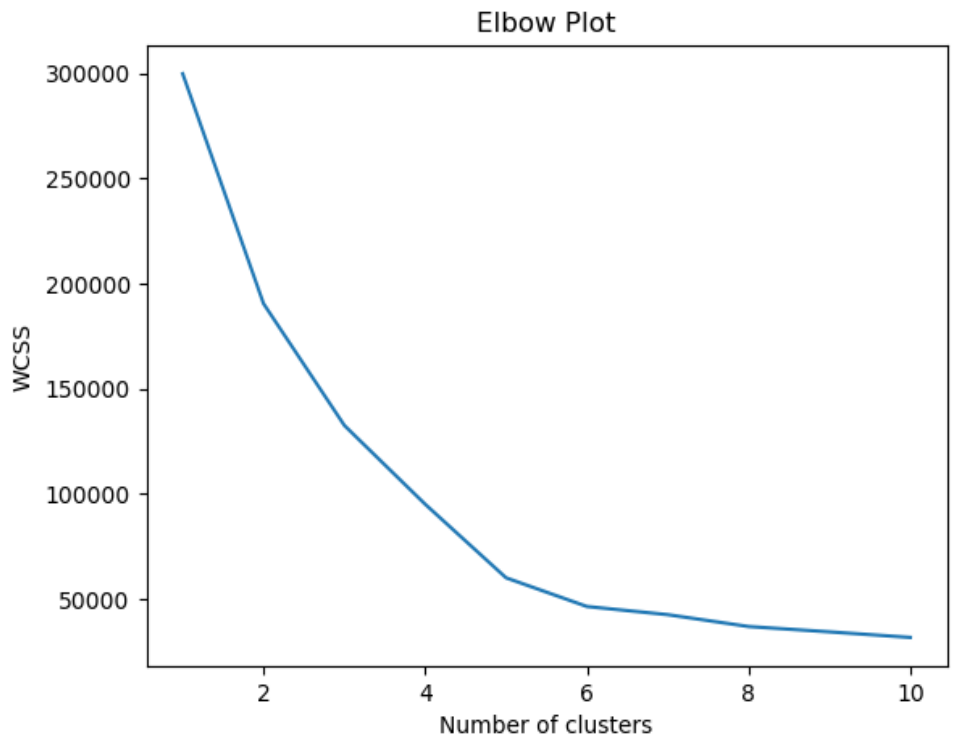


Fig. 11 Elbow plot

According to Elbow Plot, the optimum number of clusters for k means algorithm is 5.

1.7 Print silhouette scores for up to 10 clusters and identify optimum number of clusters.

```
For n_clusters=2, The Silhouette Coefficient is 0.4736752275525655
For n_clusters=3, The Silhouette Coefficient is 0.3913585278994494
For n_clusters=4, The Silhouette Coefficient is 0.45213355083408224
For n_clusters=5, The Silhouette Coefficient is 0.5565910209040732
For n_clusters=6, The Silhouette Coefficient is 0.5752600558591118
For n_clusters=7, The Silhouette Coefficient is 0.5353702011195167
For n_clusters=8, The Silhouette Coefficient is 0.46289610440630585
For n_clusters=9, The Silhouette Coefficient is 0.45849628134067916
For n_clusters=10, The Silhouette Coefficient is 0.4642198255685979
```

Fig. 12 Silhouette score

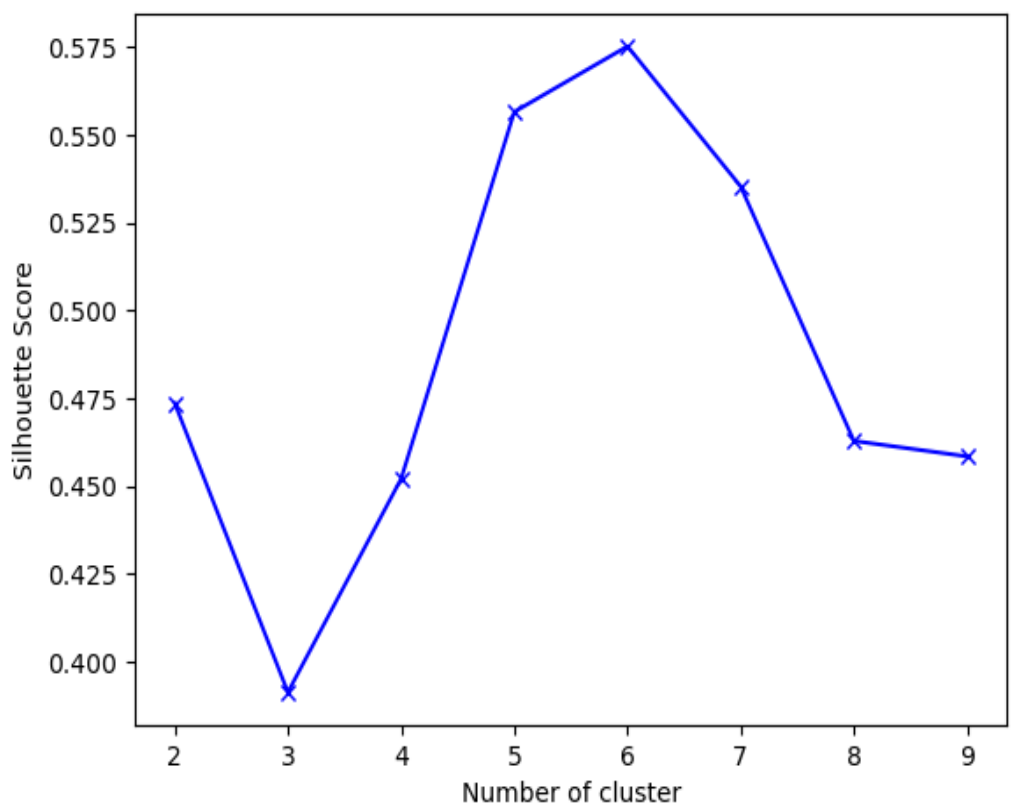


Fig. 13 Silhouette score plot

According to Silhouette Score, the optimum number of clusters for k means algorithm is 6.

1.8 Profile the ads based on optimum number of clusters using silhouette score and your domain understanding [Hint: Group the data by clusters and take sum or mean to identify trends in Clicks, spend, revenue, CPM, CTR, & CPC based on Device Type. [Make bar plots]

```
0    6842
1    1756
2    4514
3    1433
4    6994
5    1527
Name: KMeans_Labels, dtype: int64
```

Fig. 14 Proportion of data within clusters

KMeans_Labels	Ad - Length	Ad- Width	Ad Size	Available_Impressions	Matched_Queries	Impressions	Clicks	Spend	Fee	Revenue	CTR	CPM	CPC	freq
0	149.55	558.21	75690.15	46582.25	28661.60	21257.39	2947.20	318.92	0.35	208.48	15.97	14.71	0.10	6842
1	316.28	254.54	78364.78	6583616.27	3680737.02	3600777.01	8548.28	4867.49	0.32	3326.64	0.24	1.38	0.60	1756
2	695.17	316.80	213586.18	279059.43	147665.19	126758.60	13904.89	1224.16	0.35	797.23	13.64	11.92	0.09	4514
3	142.18	571.18	75625.96	843405.75	591156.63	498760.11	68157.27	7234.73	0.29	5205.55	13.77	15.13	0.11	1433
4	418.07	157.14	56445.35	2070385.26	1020575.04	980987.72	3451.11	1763.33	0.35	1157.98	0.39	1.80	0.58	6994
5	680.94	117.92	70159.76	17858169.00	9536142.81	9181756.42	17394.94	15373.73	0.24	11761.38	0.19	1.71	0.92	1527

Fig. 15 Cluster profiling

Inference

- Cluster 4 is having least value of Ad Size.
- Clusters 1 and 5 generate high Revenues.
- Available_impressions is highest for cluster 5.
- Not much difference can be seen in Fee value.

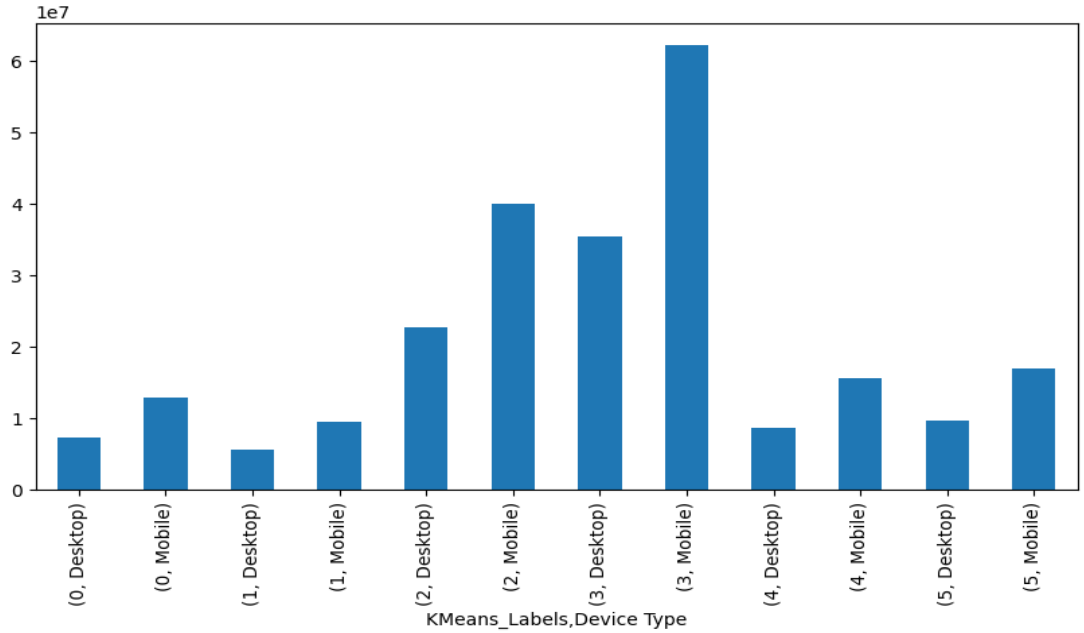


Fig. 16 Comparison of clusters according to device type and total clicks

Observation: Cluster 3 exhibits the highest click activity in the Mobile segment, followed closely by Cluster 2's Mobile segment. Cluster 3 is the only one where the desktop segment also registers a significant number of clicks.

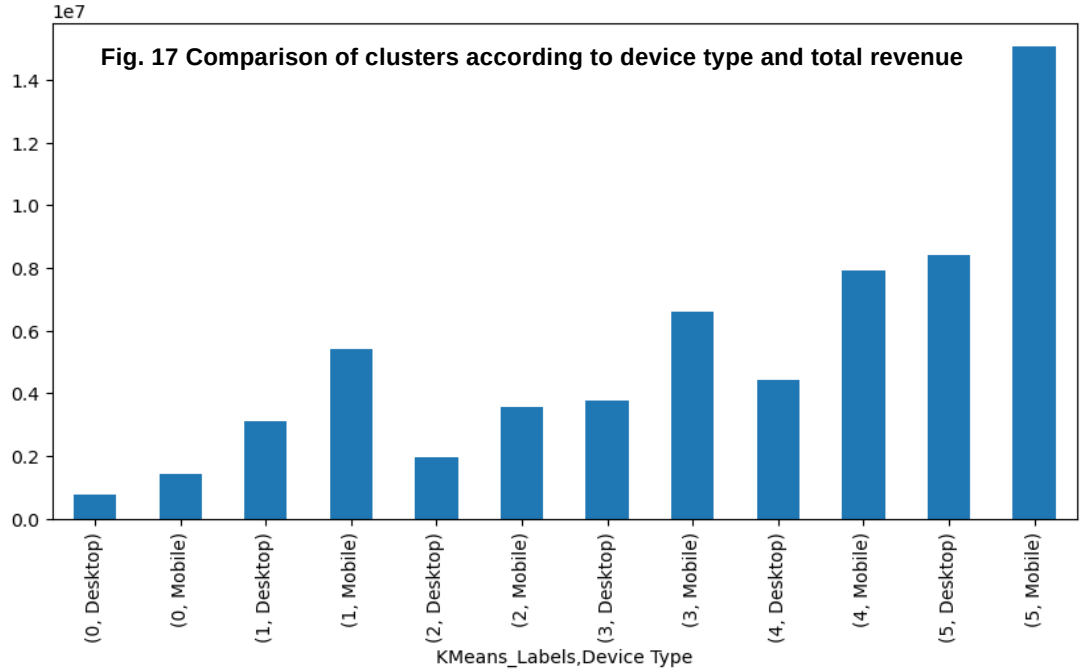


Fig. 17 Comparison of clusters according to device type and total revenue

Observation: Cluster 5 exhibits the highest revenue in the Mobile segment, contrary to Cluster 0 in the Mobile segment. Desktop segments have relatively less revenue as compared to respective clusters in Mobile segment.

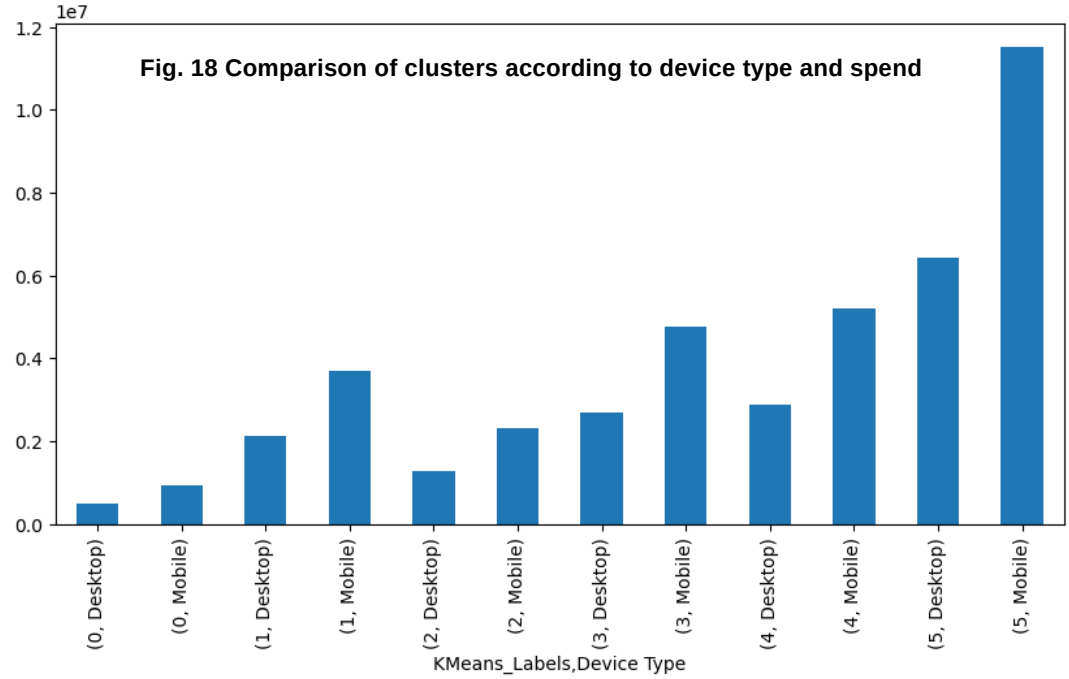


Fig. 18 Comparison of clusters according to device type and spend

Observation: The mobile segment within cluster 5 shows the highest total spending. Desktop segments have relatively less spending as compared to respective clusters in Mobile segment.

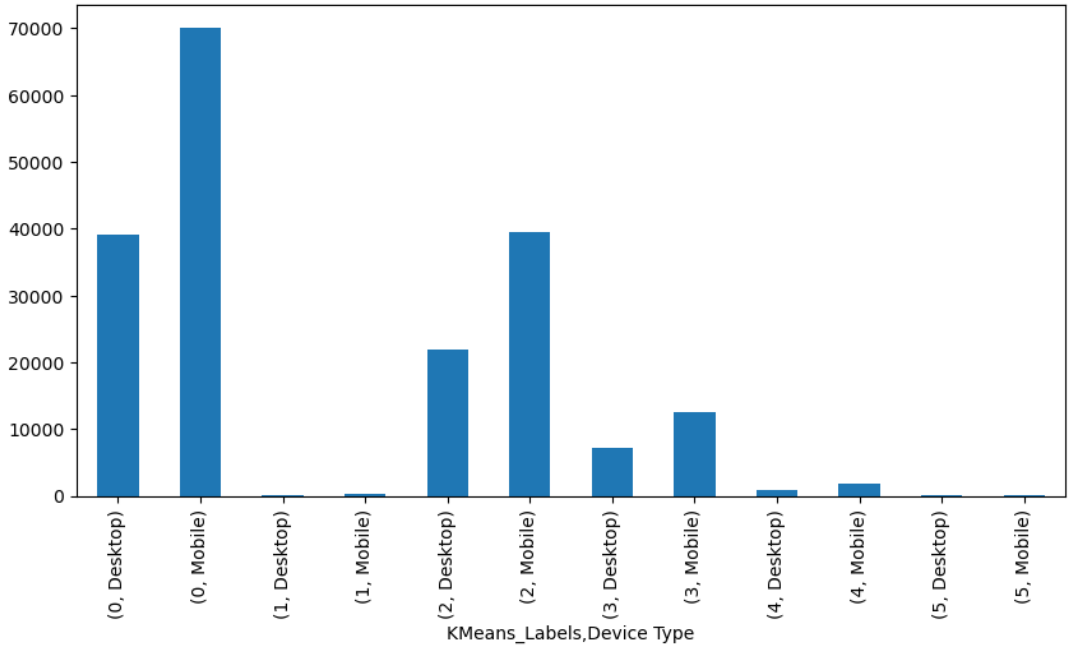
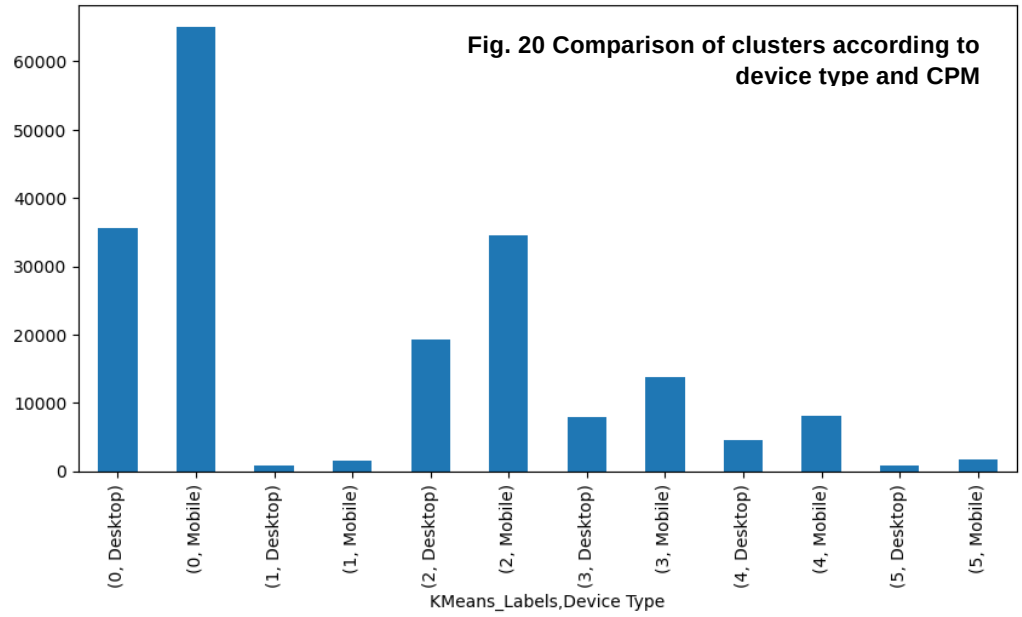
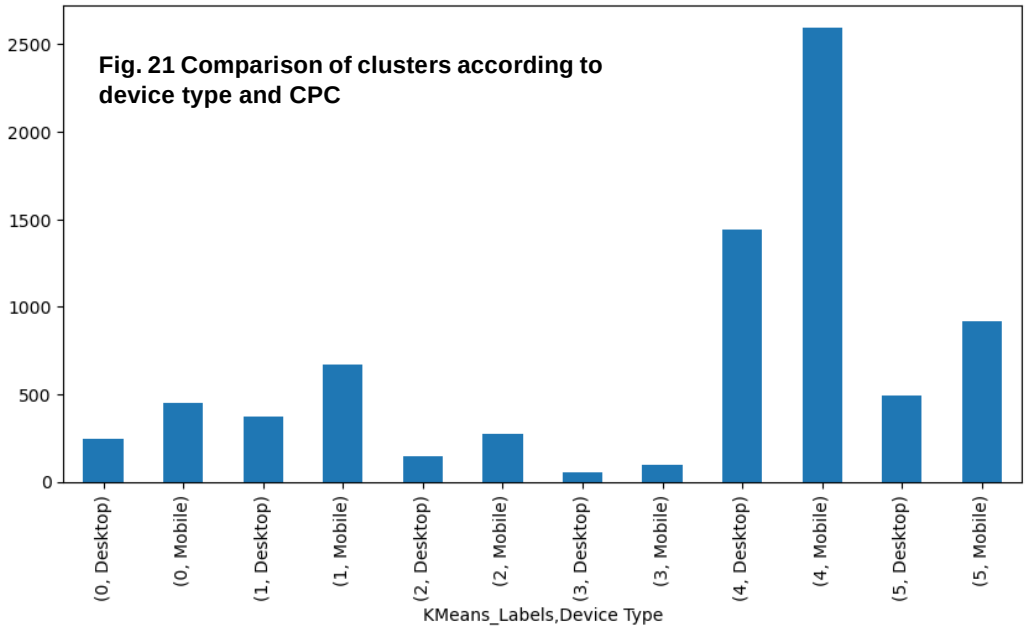


Fig. 19 Comparison of clusters according to device type and CTR

Observation: Click-through rate (CTR) is notably elevated within these same clusters. Interestingly, there appears to be no significant disparity between the mobile and desktop segments within this context.



Observation: It becomes apparent that Clusters 0, 2, and 3 exhibit the highest CPM rates. This observation suggests that these advertisements likely appear on premium and highly frequented websites.



Observation: CPC is highest for Mobile segment within Cluster 4. In this case the advertisers have significantly managed to draw attention of users. The least is shown by Desktop segment within Cluster 3.

1.9 Conclude the project by providing summary of your learnings

Based on the analyses of the digital_ads data and the clustering procedure performed, below are the key conclusions

- **Data Collection and Preprocessing**: Ads24x7, a digital marketing company, collected data to segment ads based on three essential features: CPM, CPC, and CTR. Missing values in these features were treated using a specific formula.
- **Outlier Treatment for K-Means Clustering**: Outliers can significantly impact cluster formation; hence their treatment may be necessary. The appropriate method for treating outliers can vary based on the specific dataset and goals. The available dataset is treated with IQR method.
- **Z-Score Scaling**: Z-score scaling was performed to standardize the data, which can affect the speed and performance of the K-Means algorithm positively. Thus, the data was standardized to ensure that each feature contributes equally to the clustering process.
- **Hierarchical Clustering**: Hierarchical clustering using the WARD linkage method and Euclidean distance was conducted to explore the underlying structure of the data and visualize it through a dendrogram.
- **Optimal Number of Clusters**: The elbow method and silhouette scores were utilized to determine the optimal number of clusters for K-Means clustering. These methods help identify the point at which adding more clusters does not significantly improve clustering quality.
- **Cluster Profiling**: The data was grouped by clusters, and trends in key metrics such as clicks, spend, revenue, CPM, CTR, and CPC were analyzed based on device type. Bar plots were used to visualize these trends and gain insights into cluster characteristics.

The analyses and clustering results can assist Ads24x7 in making data-driven decisions for their marketing strategies. They can use this data to targeting specific ad segments more efficiently, and optimizing their ad campaigns for better performance and ROI. Additionally, this project demonstrates the value of data analysis and clustering in enhancing marketing analytics efforts.

Problem 2

PCA

PCA FH (FT): Primary census abstract for female headed households excluding institutional households (India & States/UTs - District Level), Scheduled tribes - 2011 PCA for Female Headed Household Excluding Institutional Household. The Indian Census has the reputation of being one of the best in the world. The first Census in India was conducted in the year 1872. This was conducted at different points of time in different parts of the country. In 1881 a Census was taken for the entire country simultaneously. Since then, Census has been conducted every ten years, without a break. Thus, the Census of India 2011 was the fifteenth in this unbroken series since 1872, the seventh after independence and the second census of the third millennium and twenty first century. The census has been uninterruptedly continued despite of several adversities like wars, epidemics, natural calamities, political unrest, etc. The Census of India is conducted under the provisions of the Census Act 1948 and the Census Rules, 1990. The Primary Census Abstract which is important publication of 2011 Census gives basic information on Area, Total Number of Households, Total Population, Scheduled Castes, Scheduled Tribes Population, Population in the age group 0-6, Literates, Main Workers and Marginal Workers classified by the four broad industrial categories, namely, (i) Cultivators, (ii) Agricultural Laborers, (iii) Household Industry Workers, and (iv) Other Workers and also non-Workers.

The characteristics of the Total Population include Scheduled Castes, Scheduled Tribes, Institutional and Houseless Population and are presented by sex and rural-urban residence. Census 2011 covered 35 States/Union Territories, 640 districts, 5,924 sub-districts, 7,935 Towns and 6,40,867 Villages.

The data collected has so many variables thus making it difficult to find useful details without using Data Science Techniques. You are tasked to perform detailed EDA and identify Optimum Principal Components that explains the most variance in data. Use Sklearn only.

Note: The 24 variables given in the Rubric is just for performing EDA. You will have to consider the entire dataset, including all the variables for performing PCA.

2.1 Read the data and perform basic checks like checking head, info, summary, nulls, and duplicates, etc.

In this case study, we will examine how we can approach decisions related to Census 2011. After importing the dataset, we check the shape of the data, the types of variables and missing values in the dataset.

First five rows of the dataset are shown below.

	State Code	Dist.Code	State	Area Name	No_HH	TOT_M	TOT_F	M_06	F_06	M_SC	...	MARG_CL_0_3_M	MARG_CL_0_3_F	MARG_AL_0_3_M	MARG_AL_0_
0	1	1	Jammu & Kashmir	Kupwara	7707	23388	29796	5862	6196	3	...	1150	749	180	
1	1	2	Jammu & Kashmir	Badgam	6218	19585	23102	4482	3733	7	...	525	715	123	
2	1	3	Jammu & Kashmir	Leh(Ladakh)	4452	6546	10964	1082	1018	3	...	114	188	44	
3	1	4	Jammu & Kashmir	Kargil	1320	2784	4206	563	677	0	...	194	247	61	
4	1	5	Jammu & Kashmir	Punch	11654	20591	29981	5157	4587	20	...	874	1928	465	

5 rows × 61 columns

Fig. 22 Head of the dataset

The describe method will provide statistical information of the numerical values in the dataset like the mean, median, max. and min. values etc.

	State Code	Dist.Code	No_HH	TOT_M	TOT_F	M_06	F_06	M_SC	F_SC	M_ST	...	MAF
count	640.000000	640.000000	640.000000	640.000000	640.000000	640.000000	640.000000	640.000000	640.000000	640.000000	...	
mean	17.114062	320.500000	51222.871875	79940.576563	122372.084375	12309.098438	11942.300000	13820.946875	20778.392188	6191.807813	...	
std	9.426486	184.896367	48135.405475	73384.511114	113600.717282	11500.906881	11326.294567	14426.373130	21727.887713	9912.668948	...	
min	1.000000	1.000000	350.000000	391.000000	698.000000	56.000000	56.000000	0.000000	0.000000	0.000000	...	
25%	9.000000	160.750000	19484.000000	30228.000000	46517.750000	4733.750000	4672.250000	3466.250000	5603.250000	293.750000	...	
50%	18.000000	320.500000	35837.000000	58339.000000	87724.500000	9159.000000	8663.000000	9591.500000	13709.000000	2333.500000	...	
75%	24.000000	480.250000	68892.000000	107918.500000	164251.750000	16520.250000	15902.250000	19429.750000	29180.000000	7658.000000	...	
max	35.000000	640.000000	310450.000000	485417.000000	750392.000000	96223.000000	95129.000000	103307.000000	156429.000000	96785.000000	...	

Fig. 23 Description of the dataset

- There are 640 rows and 61 columns in the dataset.
- There are **no missing or duplicate** values.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 640 entries, 0 to 639
Data columns (total 61 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   State Code      640 non-null    int64
 1   Dist.Code       640 non-null    int64
 2   State           640 non-null    object
 3   Area Name       640 non-null    object
 4   No_HH           640 non-null    int64
 5   TOT_M           640 non-null    int64
 6   TOT_F           640 non-null    int64
 7   M_06            640 non-null    int64
 8   F_06            640 non-null    int64
 9   M_SC            640 non-null    int64
 10  F_SC            640 non-null    int64
 11  M_ST            640 non-null    int64
 12  F_ST            640 non-null    int64
 13  M_LIT           640 non-null    int64
 14  F_LIT           640 non-null    int64
 15  M_ILL           640 non-null    int64
 16  F_ILL           640 non-null    int64
 17  TOT_WORK_M      640 non-null    int64
 18  TOT_WORK_F      640 non-null    int64
 19  MAINWORK_M      640 non-null    int64
 20  MAINWORK_F      640 non-null    int64
 21  MAIN_CL_M       640 non-null    int64
 22  MAIN_CL_F       640 non-null    int64
 23  MAIN_AL_M       640 non-null    int64
 24  MAIN_AL_F       640 non-null    int64
 25  MAIN_HH_M       640 non-null    int64
 26  MAIN_HH_F       640 non-null    int64
 27  MAIN_OT_M       640 non-null    int64
 28  MAIN_OT_F       640 non-null    int64
 29  MARGWORK_M      640 non-null    int64
 30  MARGWORK_F      640 non-null    int64
 31  MARG_CL_M       640 non-null    int64
 32  MARG_CL_F       640 non-null    int64
 33  MARG_AL_M       640 non-null    int64
 34  MARG_AL_F       640 non-null    int64
 35  MARG_HH_M       640 non-null    int64
 36  MARG_HH_F       640 non-null    int64
 37  MARG_OT_M       640 non-null    int64
 38  MARG_OT_F       640 non-null    int64
 39  MARGWORK_3_6_M  640 non-null    int64
 40  MARGWORK_3_6_F  640 non-null    int64
 41  MARG_CL_3_6_M   640 non-null    int64
 42  MARG_CL_3_6_F   640 non-null    int64
 43  MARG_AL_3_6_M   640 non-null    int64
 44  MARG_AL_3_6_F   640 non-null    int64
 45  MARG_HH_3_6_M   640 non-null    int64
 46  MARG_HH_3_6_F   640 non-null    int64
 47  MARG_OT_3_6_M   640 non-null    int64
 48  MARG_OT_3_6_F   640 non-null    int64
 49  MARGWORK_0_3_M  640 non-null    int64
 50  MARGWORK_0_3_F  640 non-null    int64
 51  MARG_CL_0_3_M   640 non-null    int64
 52  MARG_CL_0_3_F   640 non-null    int64
 53  MARG_AL_0_3_M   640 non-null    int64
 54  MARG_AL_0_3_F   640 non-null    int64
 55  MARG_HH_0_3_M   640 non-null    int64
 56  MARG_HH_0_3_F   640 non-null    int64
 57  MARG_OT_0_3_M   640 non-null    int64
 58  MARG_OT_0_3_F   640 non-null    int64
 59  NON_WORK_M      640 non-null    int64
 60  NON_WORK_F      640 non-null    int64
dtypes: int64(59), object(2)
```

Fig. 24 Information of the dataset

2.2 Perform detailed Exploratory analysis by creating certain questions like (i) Which state has highest gender ratio and which has the lowest? (ii) Which district has the highest & lowest gender ratio? (Example Questions). Pick 5 variables out of the given 24 variables below for EDA: No_HH, TOT_M, TOT_F, M_06, F_06, M_SC, F_SC, M_ST, F_ST, M_LIT, F_LIT, M_ILL, F_ILL, TOT_WORK_M, TOT_WORK_F, MAINWORK_M, MAINWORK_F, MAIN_CL_M, MAIN_CL_F, MAIN_AL_M, MAIN_AL_F, MAIN_HH_M, MAIN_HH_F, MAIN_OT_M, MAIN_OT_F

- Which State has highest and lowest Total Male population?

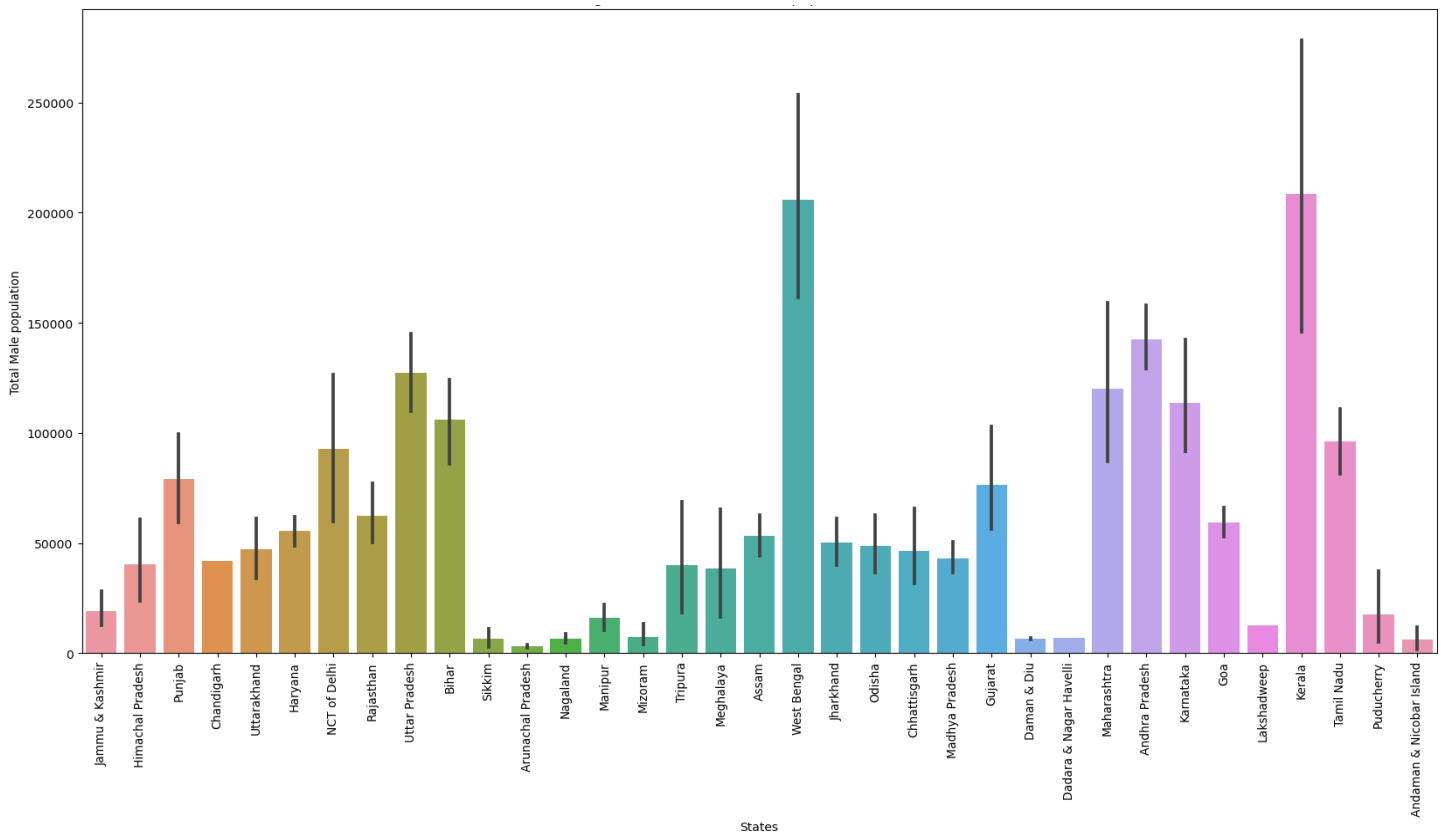


Fig. 25 Plot showing distribution of total male population in different states

Bar plot shows Kerala has highest Male population while Arunachal Pradesh has lowest.

- Which State has highest and lowest Total Female population?

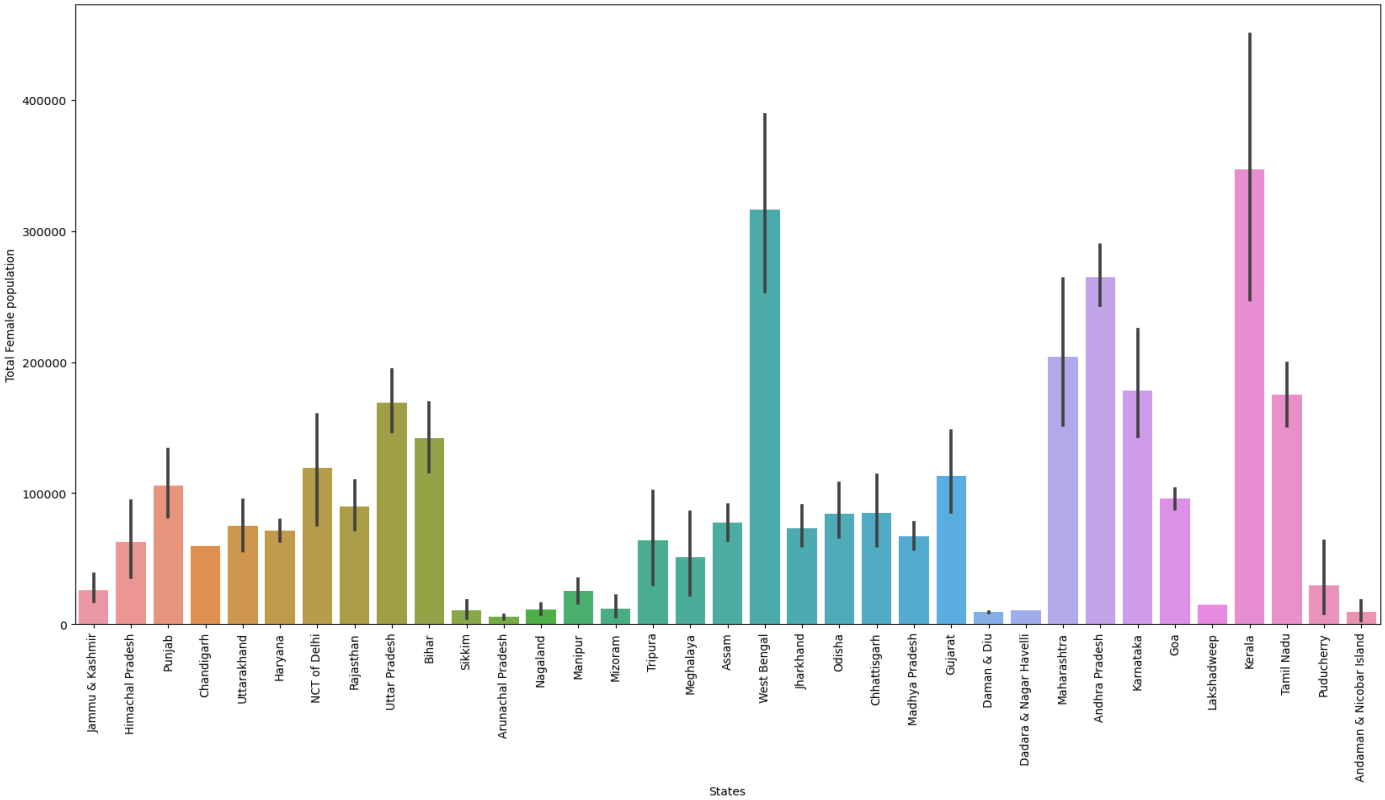


Fig. 26 Plot showing distribution of total female population in different states

Bar plot shows Kerala has highest Female population while Arunachal Pradesh has lowest.

- Which State has highest and lowest Gender Ratio in India?

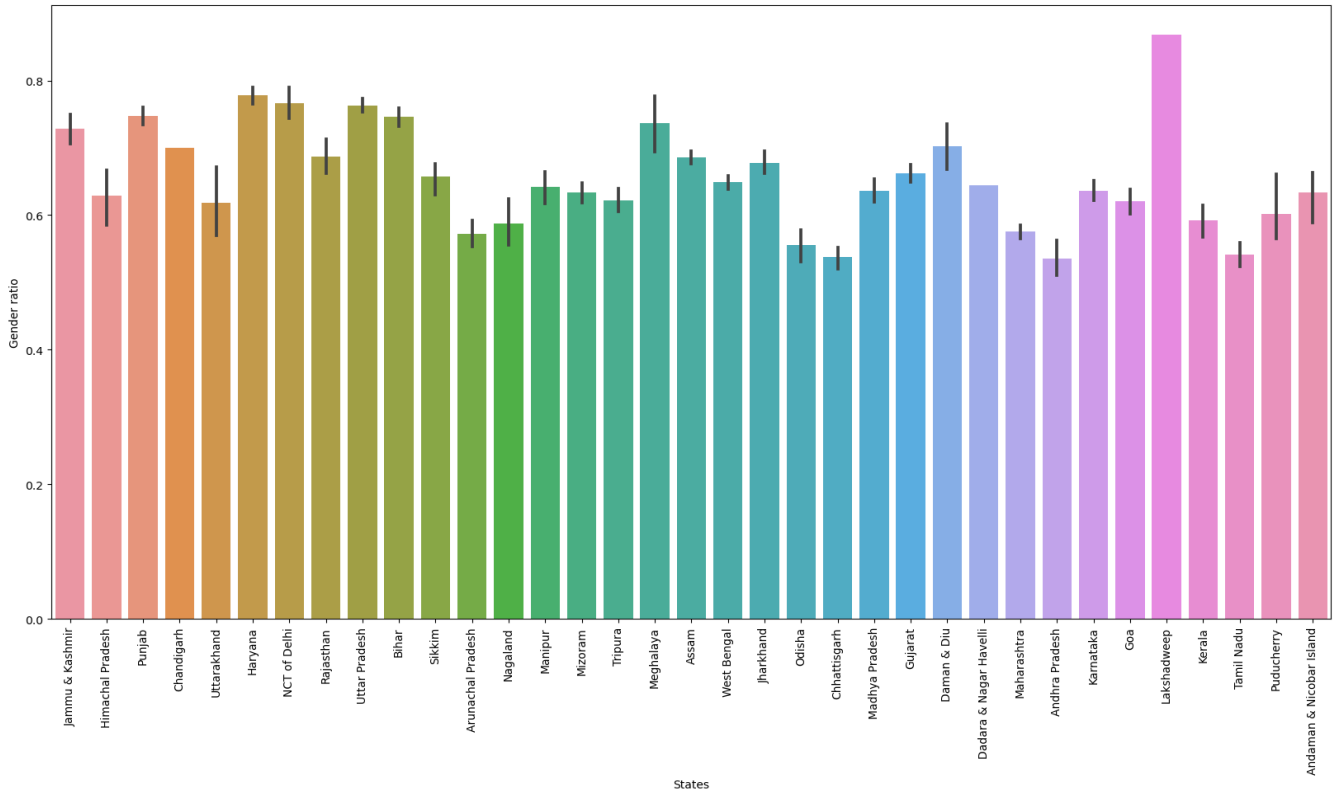


Fig. 27 Plot showing distribution of gender ratio in different states

Bar plot shows Lakshadweep has highest Gender Ratio (Male: Female) while Andra Pradesh has lowest.

- Which District has highest and lowest Gender Ratio in India?

Lakshadweep has highest Gender Ratio while Krishna has lowest.

```
Area Name
Lakshadweep       0.868061
Badgam            0.847762
Mahamaya Nagar    0.847313
Dhaulpur          0.846911
Baghpat           0.844003
                     ...
Baudh             0.451455
West Godavari     0.450076
Virudhunagar      0.449352
Koraput           0.440769
Krishna           0.437972
Name: GenderRatio, Length: 635, dtype: float64
```

Fig. 28. Distribution of gender ratio

- Which State has highest and lowest Female literate population?

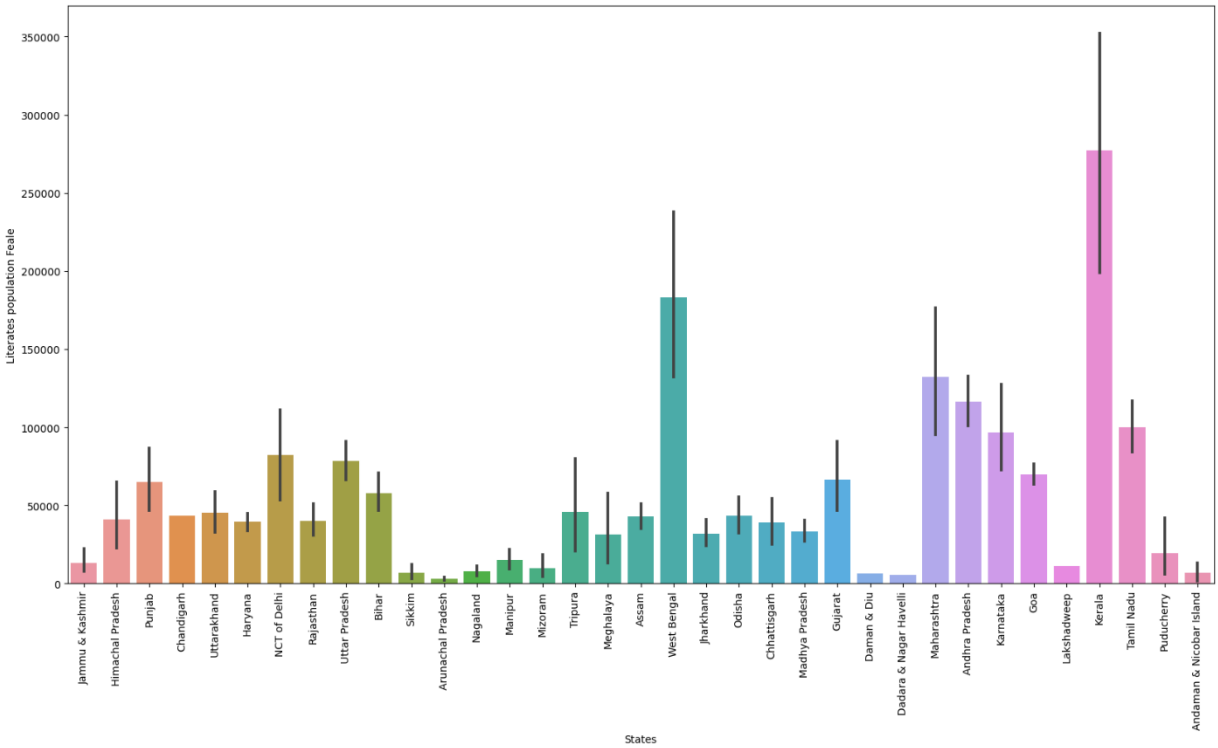


Fig. 29. Plot showing distribution of literate's population Female in different states

Bar plot shows Kerala has highest Female literate population while Arunachal Pradesh has lowest.

- Which State has highest and lowest Male literate population?

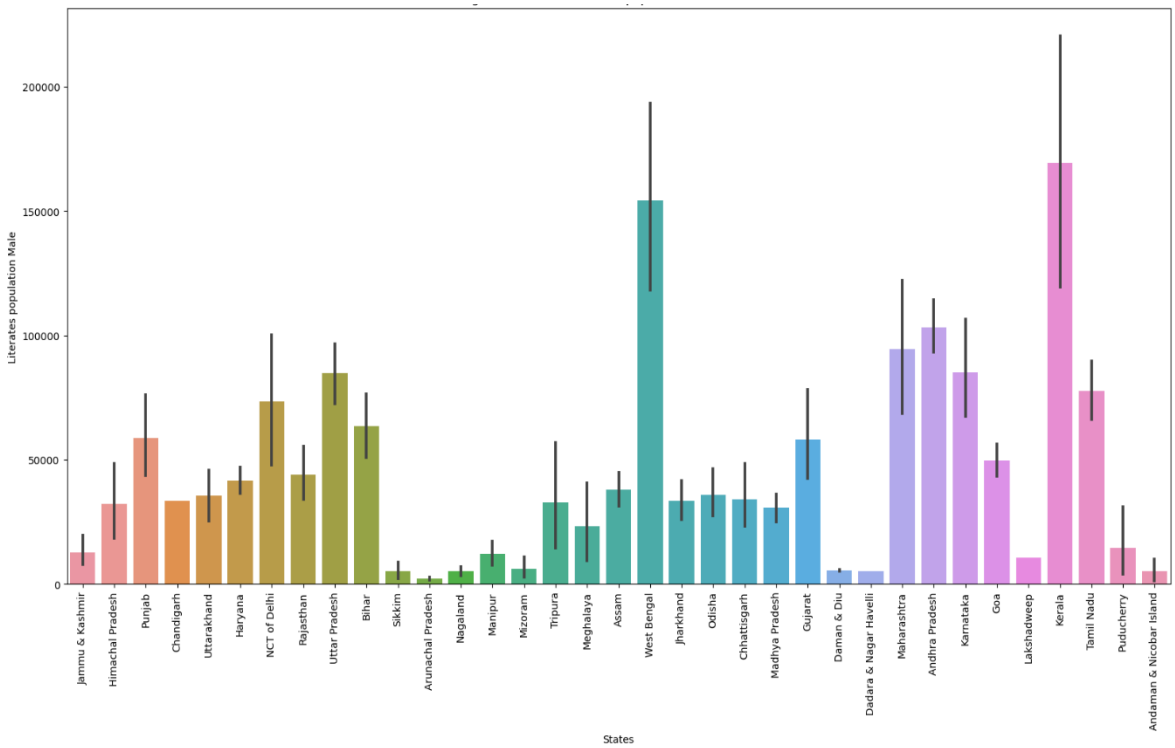


Fig. 30 Plot showing distribution of literates' population Male in different states

Bar plot shows Kerala has highest Male literate population while Arunachal Pradesh has lowest.

- Which State has highest and lowest total number of households?

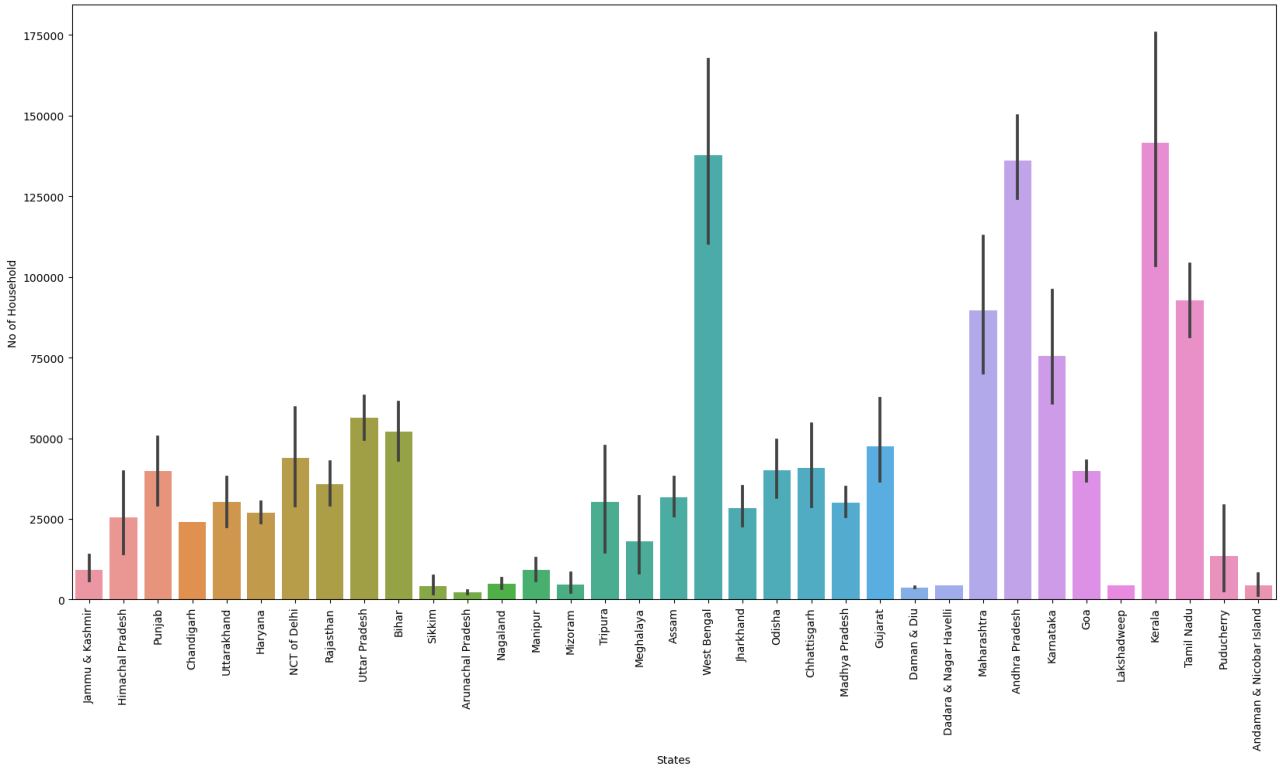
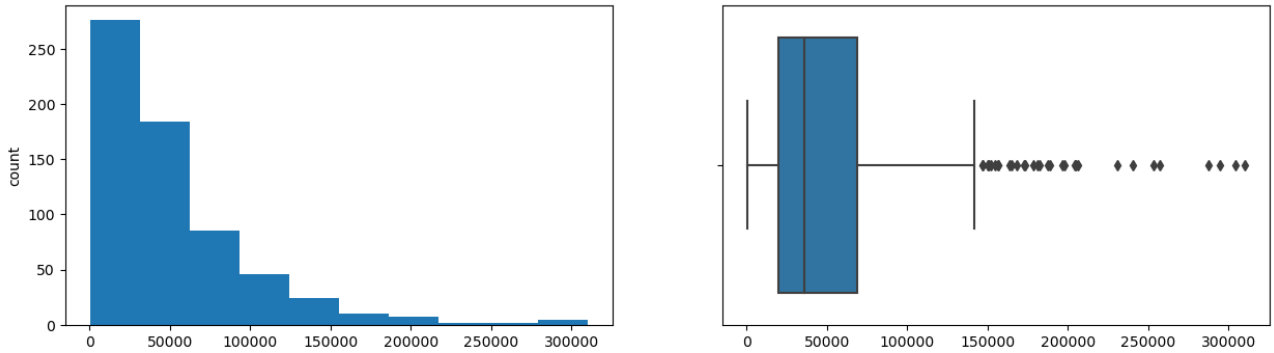


Fig. 31 Plot showing distribution of number of households in different states

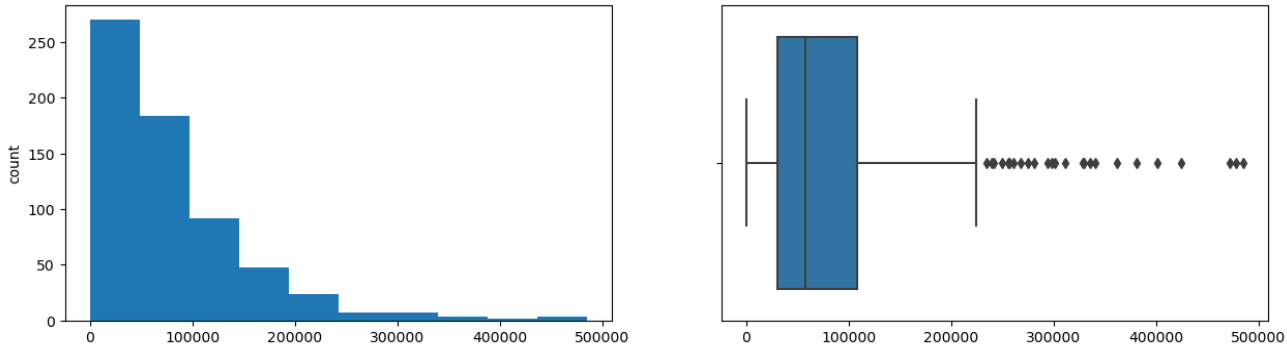
Bar plot shows Kerala has highest total number of households while Arunachal Pradesh has lowest.

Univariate Analysis-Plotting histogram and box plot

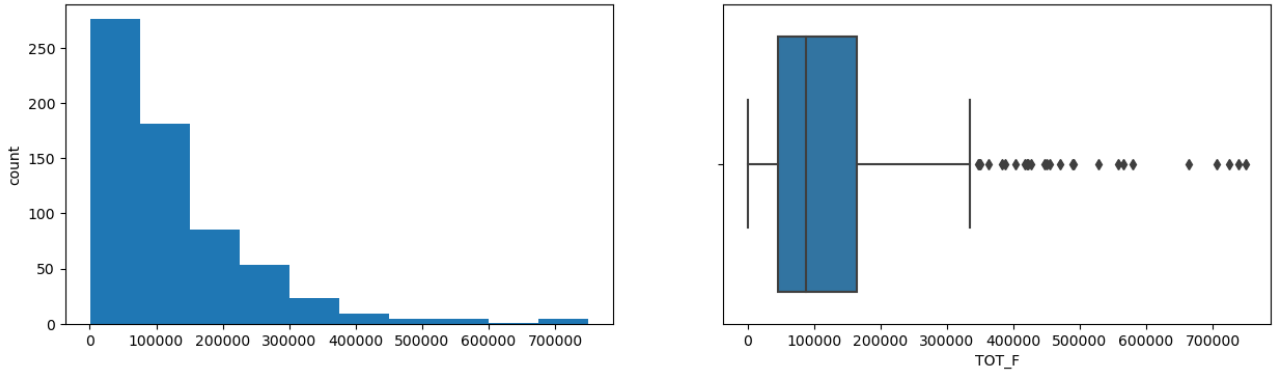
- Total number of households (No_HH)
- Total Male Population (TOT_M)
- Total Female Population (TOT_F)
- Total Female literate population(F_LIT)
- Total Male literate population(M_LIT)



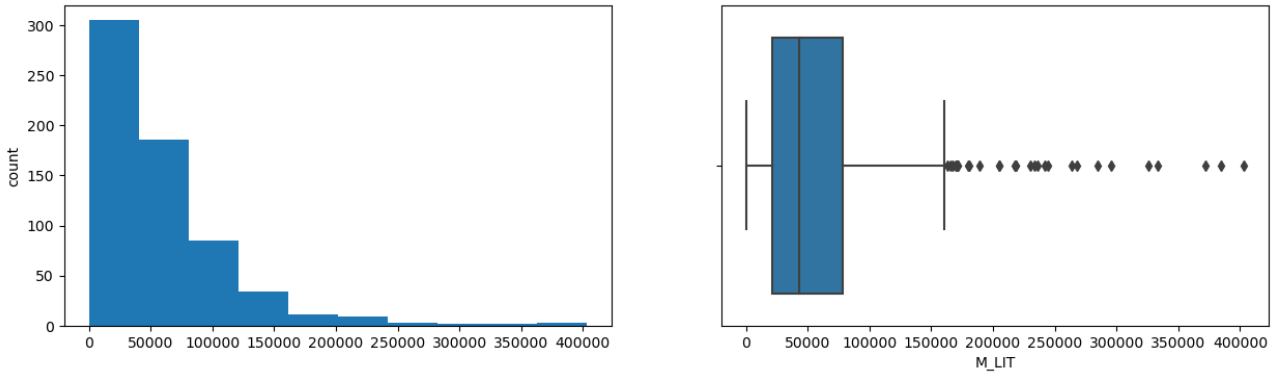
Total number of households (No_HH)



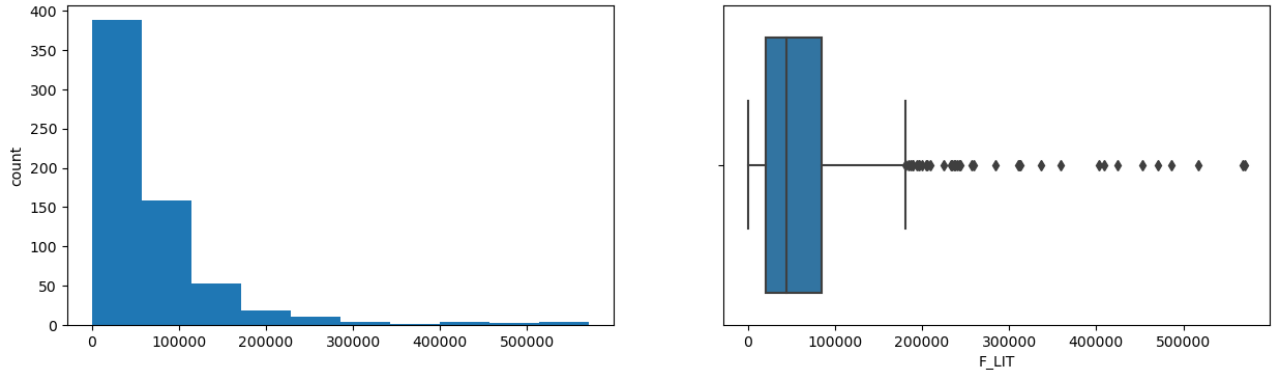
Total Male Population (TOT_M)



Total Female Population (TOT_F)



Total Female Literate Population (F_LIT)



Total Male Literate Population (F_LIT)

Fig. 32 Box plot showing univariate analysis of different components.

From the Univariate analysis it can be seen that all variables are Left skewed here and all are having Outliers.

Bivariate Analysis-Plotting heatmap

From the Bivariate Analysis it can be seen that majority of the variables are positively correlated to each other.

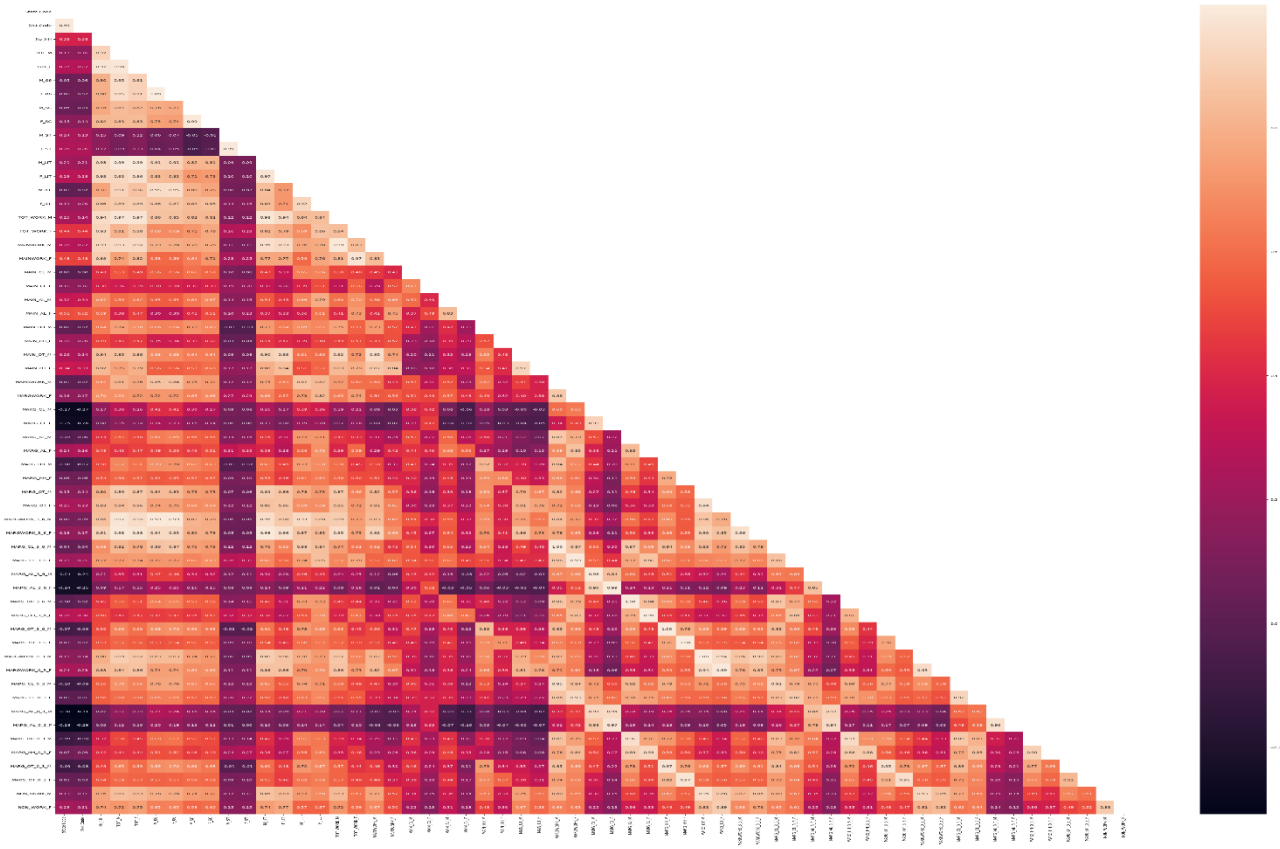


Fig. 33 Heat map showing bivariate analysis of different components.

2.3 We choose not to treat outliers for this case. Do you think that treating outliers for this case is necessary?

The figure shows number of outliers in different columns of the dataset. It can be seen that the outlier percentage is around 0.04-0.09%, which is not significant.

Hence it is not required to treat outliers in this case.

No_HH	31
TOT_M	25
TOT_F	26
M_06	32
F_06	33
M_SC	29
F_SC	29
M_ST	51
F_ST	58
M_LIT	30
F_LIT	37
M_ILL	39
F_ILL	26
TOT_WORK_M	32
TOT_WORK_F	42
MAINWORK_M	36
MAINWORK_F	55
MAIN_CL_M	25
MAIN_CL_F	29
MAIN_AL_M	36
MAIN_AL_F	60
MAIN_HH_M	47
MAIN_HH_F	56
MAIN_OT_M	53
MAIN_OT_F	59
MARGWORK_M	43
MARGWORK_F	19
MARG_CL_M	55
MARG_CL_F	53
MARG_AL_M	48
MARG_AL_F	30
MARG_HH_M	58
MARG_HH_F	39
MARG_OT_M	46
MARG_OT_F	42
MARGWORK_3_6_M	31
MARGWORK_3_6_F	32
MARG_CL_3_6_M	44
MARG_CL_3_6_F	15
MARG_AL_3_6_M	56
MARG_AL_3_6_F	49
MARG_HH_3_6_M	43
MARG_HH_3_6_F	25
MARG_OT_3_6_M	58
MARG_OT_3_6_F	40
MARGWORK_0_3_M	48
MARGWORK_0_3_F	40
MARG_CL_0_3_M	48
MARG_CL_0_3_F	35
MARG_AL_0_3_M	55
MARG_AL_0_3_F	59
MARG_HH_0_3_M	63
MARG_HH_0_3_F	46
MARG_OT_0_3_M	60
MARG_OT_0_3_F	42
NON_WORK_M	54
NON_WORK_F	47
dtype: int64	

Fig. 34 Proportion of outliers across dataset

2.4 Scale the Data using z-score method. Does scaling have any impact on outliers? Compare boxplots before and after scaling and comment.

A separate dataset is created after dropping 4 columns 'State Code', 'Dist.Code', 'State', 'Area Name'.

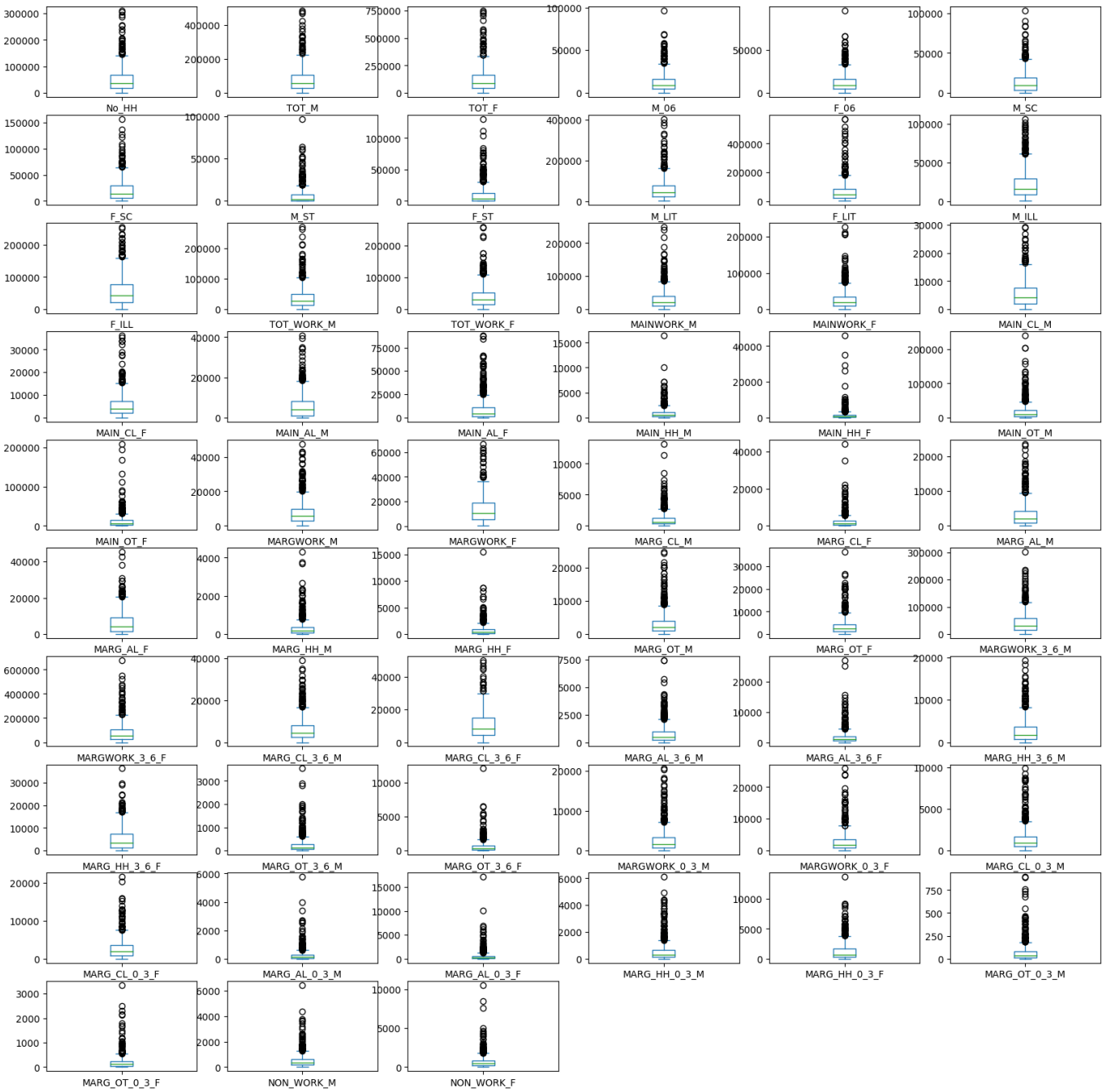


Fig. 35 Box plot for outliers before scaling

After scaling the below dataset is obtained

	No_HH	TOT_M	TOT_F	M_06	F_06	M_SC	F_SC	M_ST	F_ST	M_LIT	...	MARG_CL_0_3_M	MARG_CL_0_3_F
0	-0.904738	-0.771236	-0.815563	-0.561012	-0.507738	-0.958575	-0.957049	-0.423306	-0.476423	-0.798097	...	-0.163229	-0.720610
1	-0.935695	-0.823100	-0.874534	-0.681096	-0.725367	-0.958297	-0.956772	-0.582014	-0.607607	-0.849434	...	-0.583103	-0.732811
2	-0.972412	-1.000919	-0.981466	-0.976956	-0.965262	-0.958575	-0.956772	-0.038951	-0.027273	-0.956457	...	-0.859212	-0.921931
3	-1.037530	-1.052224	-1.041001	-1.022118	-0.995393	-0.958783	-0.957049	-0.355965	-0.390060	-1.004643	...	-0.805468	-0.900758
4	-0.822676	-0.809381	-0.813933	-0.622359	-0.649908	-0.957395	-0.955529	0.149238	0.043330	-0.800568	...	-0.348645	-0.297513

Fig. 36 Scaled dataset

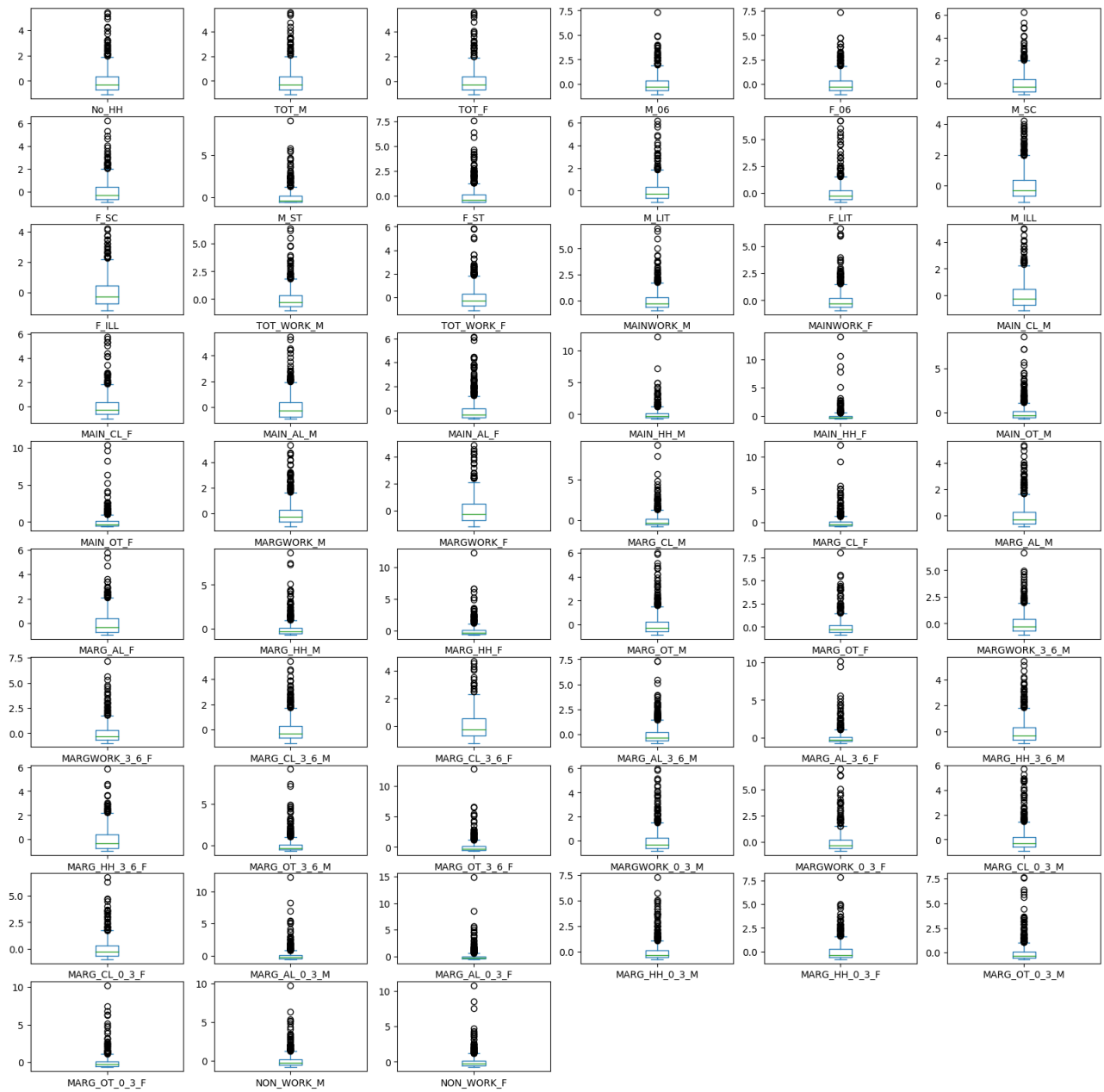


Fig. 37 Box plot for outliers after scaling

The boxplots obtained after scaling the dataset, shows no variation from original dataset, thereby indicating that there is no effect of scaling on outliers.

2.5 Perform all the required steps for PCA (use Sklearn only) Create the covariance Matrix Get eigen values and eigen vector.

After successfully performing Bartlett Sphericity and KMO Test, we can go ahead with Principal Component Analysis (PCA) following the steps below:

- Performing Outlier Treatment
- Scaling of the data
- Create Covariance Matrix
- Extract Eigen Vectors
- Find Eigen Value
- Create WSS Scree plot for variance
- Find a cut off for selecting the number of PCs

We have imported PCA from sklearn.decomposition and obtained covariance Matrix, eigen values and eigen vectors.

```
array([[1.00156495, 0.91760364, 0.97210871, ..., 0.53769433, 0.76357722,
        0.73684378],
       [0.91760364, 1.00156495, 0.98417823, ..., 0.5891007 , 0.84621844,
        0.71718181],
       [0.97210871, 0.98417823, 1.00156495, ..., 0.572748  , 0.82894851,
        0.74775097],
       ...,
       [0.53769433, 0.5891007 , 0.572748  , ..., 1.00156495, 0.61052325,
        0.52191235],
       [0.76357722, 0.84621844, 0.82894851, ..., 0.61052325, 1.00156495,
        0.88228018],
       [0.73684378, 0.71718181, 0.74775097, ..., 0.52191235, 0.88228018,
        1.00156495]])
```

Fig. 38 Covariance matrix

```
array([[ 0.15602058,  0.16711763,  0.16555318, ...,  0.13219224,
         0.15037558,  0.1310662 ],
       [-0.12634653, -0.08967655, -0.10491237, ...,  0.05081332,
        -0.06536455, -0.07384742],
       [-0.00269025,  0.05669762,  0.03874947, ..., -0.07871987,
         0.11182732,  0.1025525 ],
       ...,
       [ 0.        ,  0.2077636 ,  0.24647657, ..., -0.07217993,
         0.00399206, -0.06929081],
       [ 0.        ,  0.2887035 , -0.20596721, ...,  0.04019745,
        -0.03192722,  0.00778048],
       [-0.        ,  0.18790022,  0.02642675, ..., -0.02597314,
        -0.13972835, -0.02147533]])
```

Fig. 39. Eigen vectors

```
array([3.18135647e+01, 7.86942415e+00, 4.15340812e+00, 3.66879058e+00,
       2.20652588e+00, 1.93827502e+00, 1.17617374e+00, 7.51159086e-01,
       6.17053743e-01, 5.28300887e-01, 4.29831189e-01, 3.53440201e-01,
       2.96163013e-01, 2.81275560e-01, 1.92158325e-01, 1.36267920e-01,
       1.13389199e-01, 1.06303946e-01, 9.72885376e-02, 8.01062194e-02,
       5.76089954e-02, 4.43955966e-02, 3.78910846e-02, 2.96360194e-02,
       2.70797618e-02, 2.34458139e-02, 1.45111511e-02, 1.09852268e-02,
       9.31507853e-03, 8.13540203e-03, 7.89250253e-03, 5.02601514e-03,
       2.59771182e-03, 1.06789820e-03, 7.13559124e-04, 2.47799812e-31,
       2.47799812e-31, 2.47799812e-31, 2.47799812e-31, 2.47799812e-31,
       2.47799812e-31, 2.47799812e-31, 2.47799812e-31, 2.47799812e-31,
       2.47799812e-31, 2.47799812e-31, 2.47799812e-31, 2.47799812e-31,
       2.47799812e-31, 2.47799812e-31, 2.47799812e-31, 2.47799812e-31,
       2.47799812e-31, 2.47799812e-31, 2.47799812e-31, 2.47799812e-31,
       2.47799812e-31])
```

Fig. 40 Eigen values

2.6 Identify the optimum number of PCs (for this project, take at least 90% explained variance). Show Scree plot.

Explained variance = (eigen value of each PC)/ (sum of eigen values of all PCs)

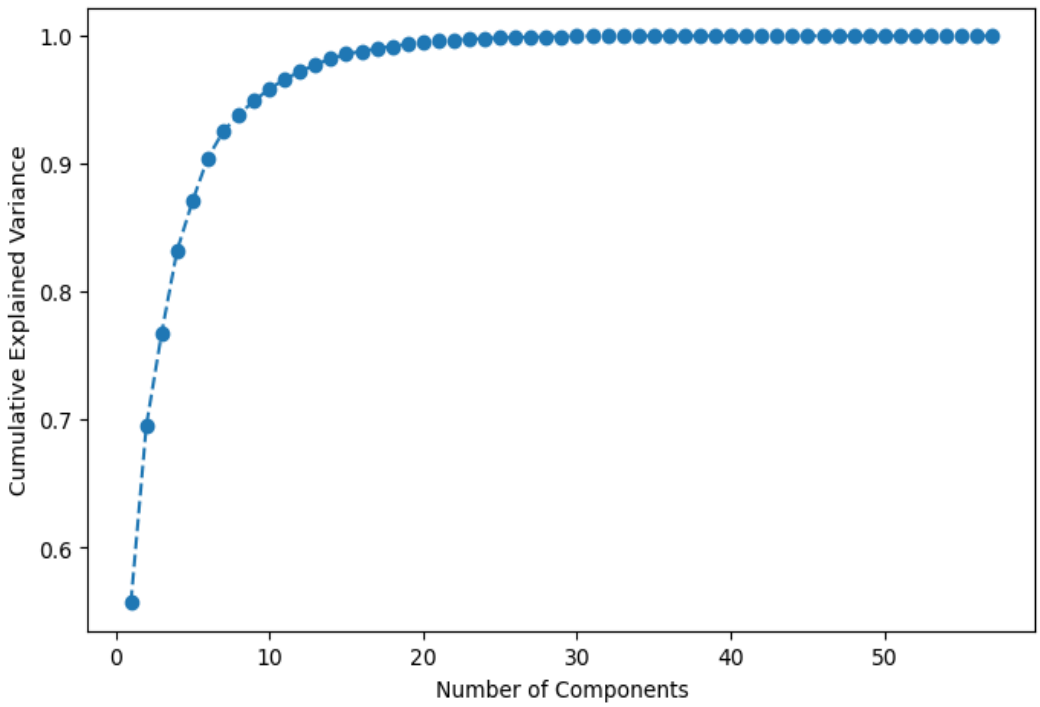


Fig. 41 Explained variances by components

```
Cumulative Variance Explained in Percentage: [ 55.73  69.51  76.79  83.21  87.08  90.47  92.53  93.85  94.93  95.85
  96.61  97.23  97.75  98.24  98.57  98.81  99.01  99.2   99.37  99.51
  99.61  99.69  99.75  99.81  99.85  99.89  99.92  99.94  99.96  99.97
  99.98  99.99 100.   100.   100.   100.   100.   100.   100.   100.
 100.   100.   100.   100.   100.   100.   100.   100.   100.   100.
 100.   100.   100.   100.   100.   100.   100.  ]
```

Fig. 42 Cumulative variance explained in percentage

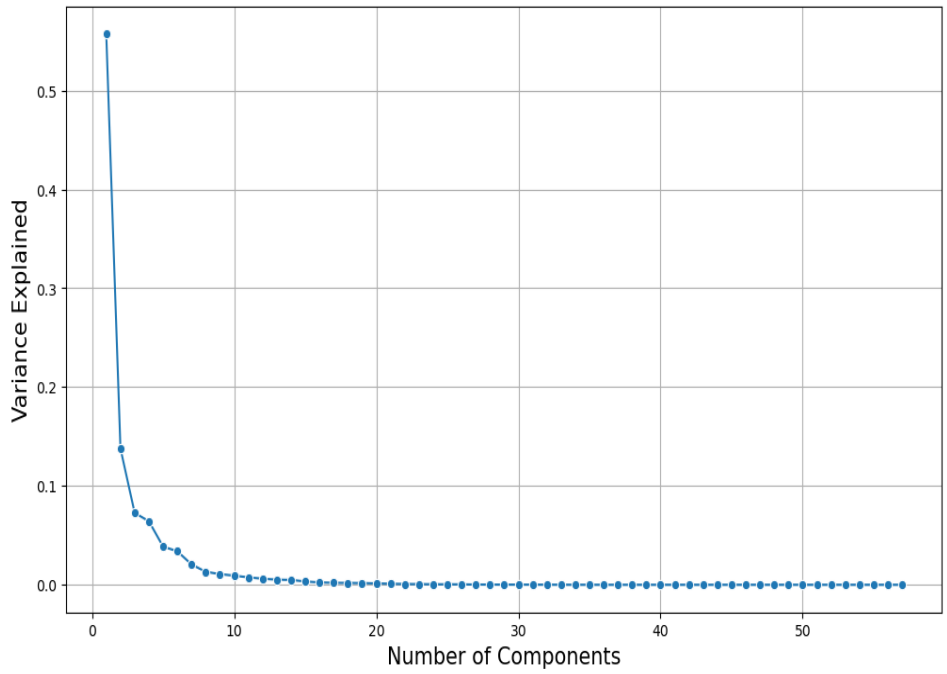


Fig. 43 Scree Plot

The number of components can be decided based upon the explained variance. Here, it is decided to keep the number of components as 6 as the cumulative explained variance is around 90%

2.7 Compare PCs with Actual Columns and identify which is explaining most variance. Write inferences about all the principal components in terms of actual variables.

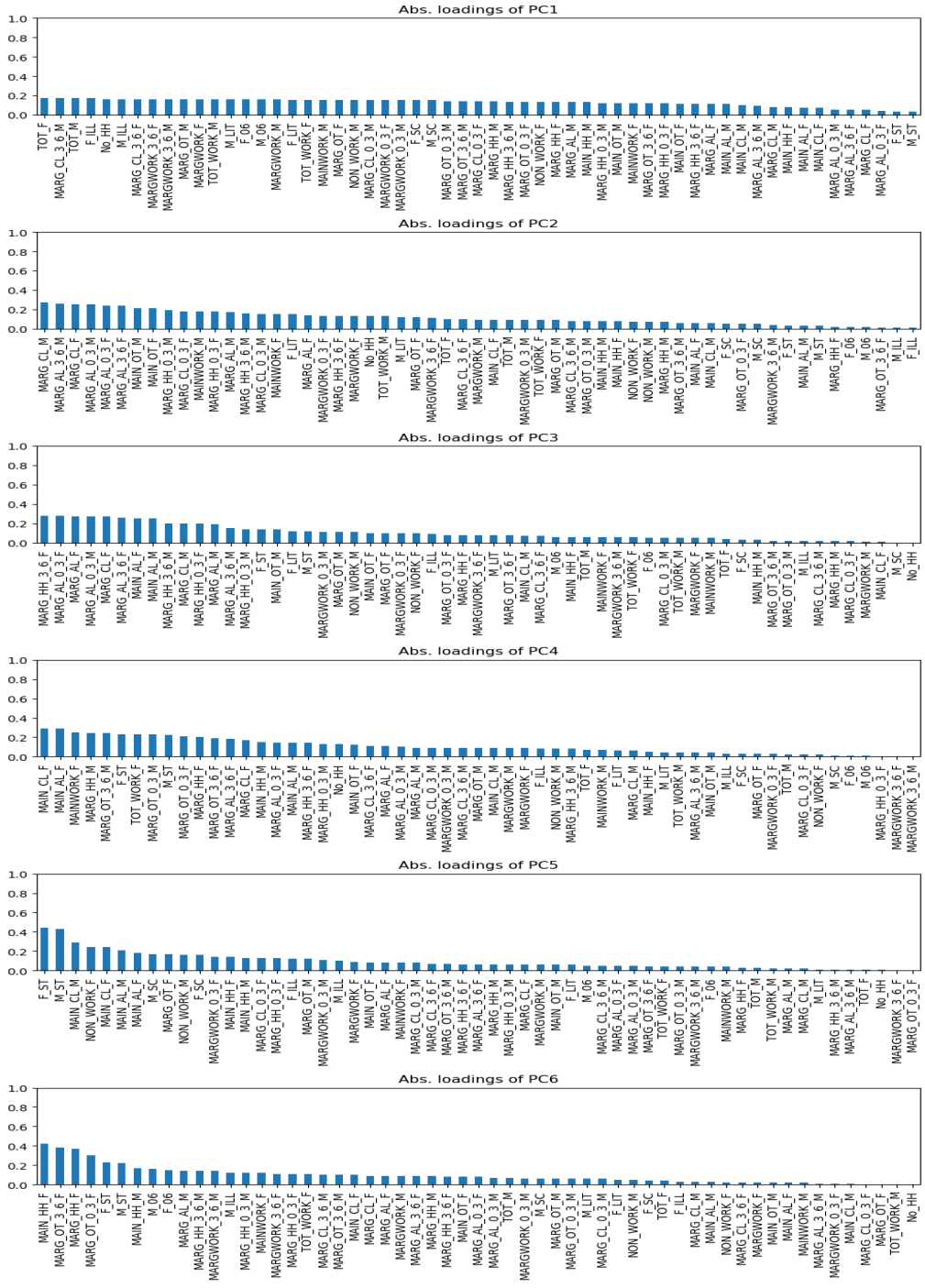


Fig. 44 Comparison of original variables with different PCs

PC1 is explaining most variance.

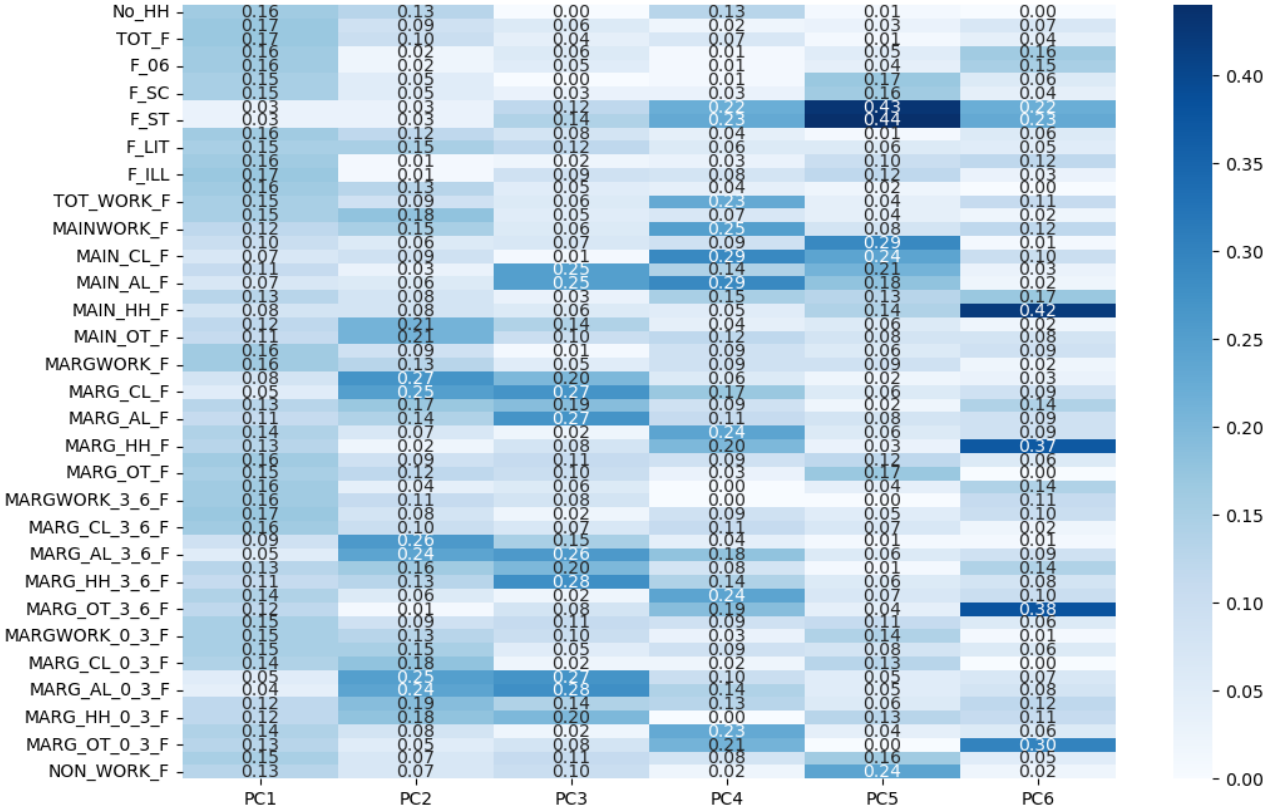


Fig. 45 Influence of original variables on different PCs

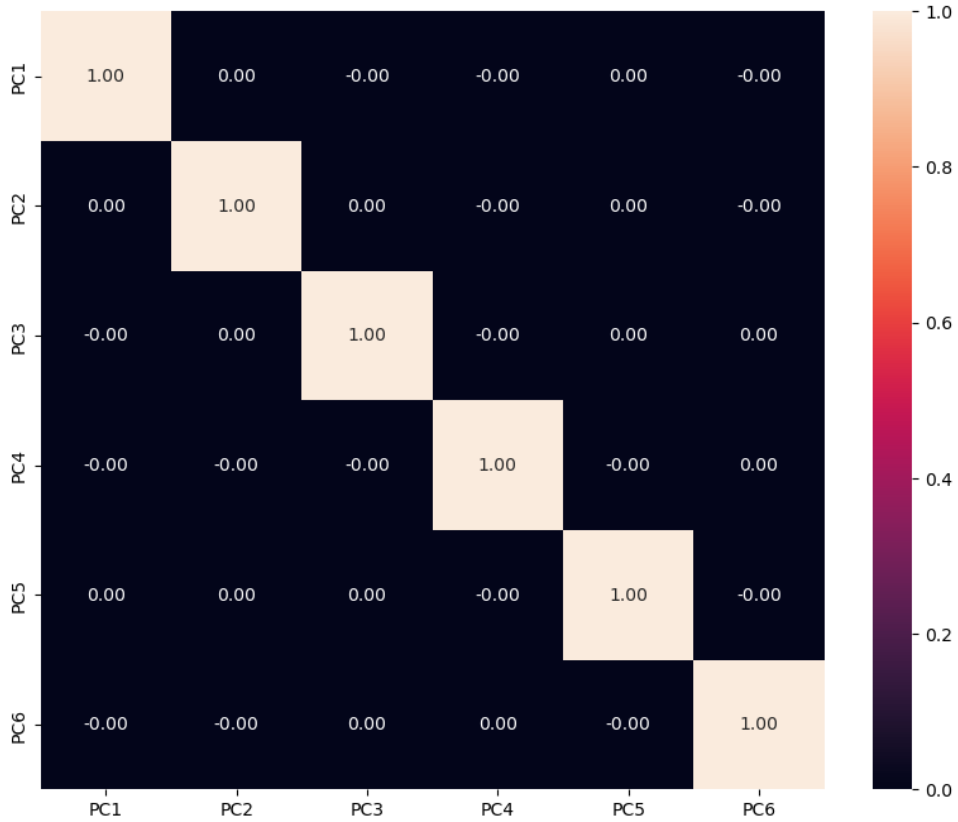


Fig. 46 Correlation between different PCs

2.8 Write linear equation for first PC.

```
( 0.16 ) * No_HH + ( 0.17 ) * TOT_M + ( 0.17 ) * TOT_F + ( 0.16 ) * M_06 + ( 0.16 ) * F_06 + ( 0.15 ) * M_SC + ( 0.15 ) * F_SC
+ ( 0.03 ) * M_ST + ( 0.03 ) * F_ST + ( 0.16 ) * M_LIT + ( 0.15 ) * F_LIT + ( 0.16 ) * M_ILL + ( 0.17 ) * F_ILL + ( 0.16 ) * TO
T_WORK_M + ( 0.15 ) * TOT_WORK_F + ( 0.15 ) * MAINWORK_M + ( 0.12 ) * MAINWORK_F + ( 0.1 ) * MAIN_CL_M + ( 0.07 ) * MAIN_CL_F +
( 0.11 ) * MAIN_AL_M + ( 0.07 ) * MAIN_AL_F + ( 0.13 ) * MAIN_HH_M + ( 0.08 ) * MAIN_HH_F + ( 0.12 ) * MAIN_OT_M + ( 0.11 ) * M
AIN_OT_F + ( 0.16 ) * MARGWORK_M + ( 0.16 ) * MARGWORK_F + ( 0.08 ) * MARG_CL_M + ( 0.05 ) * MARG_CL_F + ( 0.13 ) * MARG_AL_M +
( 0.11 ) * MARG_AL_F + ( 0.14 ) * MARG_HH_M + ( 0.13 ) * MARG_HH_F + ( 0.16 ) * MARG_OT_M + ( 0.15 ) * MARG_OT_F + ( 0.16 ) * M
ARGWORK_3_6_M + ( 0.16 ) * MARGWORK_3_6_F + ( 0.17 ) * MARG_CL_3_6_M + ( 0.16 ) * MARG_CL_3_6_F + ( 0.09 ) * MARG_AL_3_6_M + (
0.05 ) * MARG_AL_3_6_F + ( 0.13 ) * MARG_HH_3_6_M + ( 0.11 ) * MARG_HH_3_6_F + ( 0.14 ) * MARG_OT_3_6_M + ( 0.12 ) * MARG_OT_3_
6_F + ( 0.15 ) * MARGWORK_0_3_M + ( 0.15 ) * MARGWORK_0_3_F + ( 0.15 ) * MARG_CL_0_3_M + ( 0.14 ) * MARG_CL_0_3_F + ( 0.05 ) *
MARG_AL_0_3_M + ( 0.04 ) * MARG_AL_0_3_F + ( 0.12 ) * MARG_HH_0_3_M + ( 0.12 ) * MARG_HH_0_3_F + ( 0.14 ) * MARG_OT_0_3_M + (
0.13 ) * MARG_OT_0_3_F + ( 0.15 ) * NON_WORK_M + ( 0.13 ) * NON_WORK_F +
```

Fig. 47 Linear equation for first PC